2 Unit-II: Limits and Continuity

Limits: Introduction

Consider the function

$$f(x) = \frac{x^2 - 4}{x - 2}, \qquad x \neq 2.$$

At $x = 2$, the expression is not defined because the denominator becomes zero. However, for $x \neq 2$,

$$x^2 - 4 = (x - 2)(x + 2),$$

and hence

$$f(x) = \frac{(x - 2)(x + 2)}{x - 2} = x + 2.$$

Now let us examine some values of $f(x)$ as x approaches 2.

x	$f(x)$
1.9	3.9
1.99	3.99
1.999	3.999
2.001	4.001
2.01	4.01
2.1	4.1

From the table, we observe that as x gets closer and closer to 2, the values of $f(x)$ get closer and closer to 4.

Therefore,

$$\boxed{\lim_{x \to 2} \frac{x^2 - 4}{x - 2} = 4}.$$

Even though the original function is not defined at $x = 2$, its limit exists and equals 4.

This shows that the limit depends on the behavior of the function near the point and not necessarily on the value of the function at the point.

The idea of a *limit* describes the value that a function approaches when the independent variable approaches a particular point.

Let f be a real-valued function defined in a neighborhood of a, except possibly at a itself. We say that

$$\lim_{x \to a} f(x) = L$$

if the values of $f(x)$ can be made arbitrarily close to L by taking x sufficiently close to a, with $x \neq a$.

Notice that the value $f(a)$ is not important for the existence of the limit. The limit concerns the behavior of $f(x)$ *near* a.

Definition, Meaning and Geometrical Interpretation

Definition 2.1 (ε-δ **Definition).** We say that

$$\boxed{\lim_{x\to a} f(x) = L}$$

if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that $|f(x)-L| < \varepsilon$ whenever $0 < |x-a| < \delta$
Symbolically, $\boxed{\forall \varepsilon > 0\ \exists \delta > 0\ \ni 0 < |x-a| < \delta \Longrightarrow |f(x)-L| < \varepsilon}$

This is the rigorous definition of the limit of a function.

Meaning of ε:

The Greek letter

$$\varepsilon > 0$$

measures how close we want $f(x)$ to be to the proposed limit L.

The inequality

$$|f(x) - L| < \varepsilon$$

means that the distance between $f(x)$ and L is less than ε.

Equivalently,

$$L - \varepsilon < f(x) < L + \varepsilon.$$

Thus $f(x)$ lies inside the interval $\boxed{(L-\varepsilon, L+\varepsilon)}$.

This interval is called an ε-neighborhood of L.

Meaning of δ:

The number

$$\delta > 0$$

controls how close x must be to a.

The inequality

$$|x - a| < \delta$$

is equivalent to

$$a - \delta < x < a + \delta.$$

Because we exclude $x = a$, we write

$$0 < |x - a| < \delta.$$

Thus x lies in the deleted neighborhood

$$(a - \delta, a) \cup (a, a + \delta).$$

Geometrical Interpretation:

The definition

$$0 < |x - a| < \delta \quad \Longrightarrow \quad |f(x) - L| < \varepsilon$$

can be understood using two bands.

The condition

$$|x - a| < \delta$$

gives the vertical strip

$$a - \delta < x < a + \delta.$$

The condition

$$|f(x) - L| < \varepsilon$$

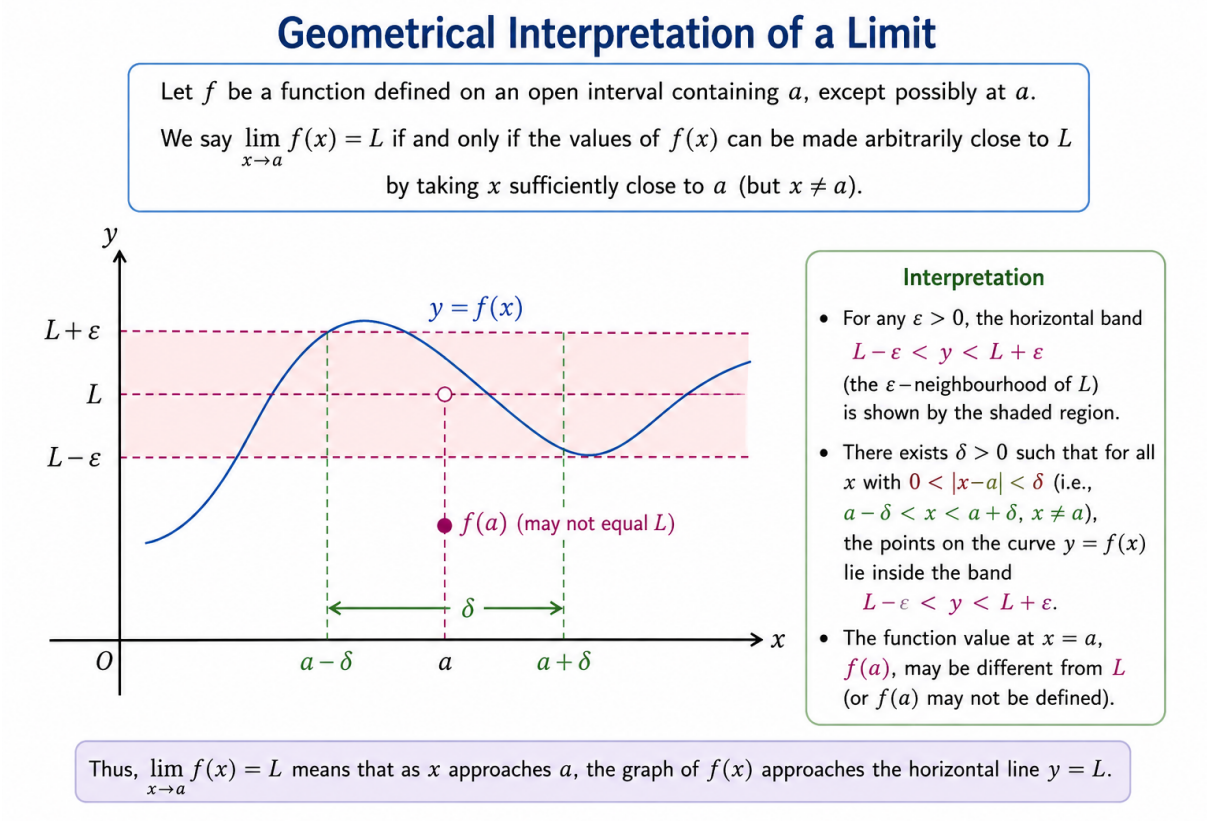
gives the horizontal strip

$$L - \varepsilon < y < L + \varepsilon.$$

Thus,

$$\lim_{x \to a} f(x) = L$$

means that, no matter how narrow a horizontal strip around L is chosen, we can find a sufficiently narrow vertical strip around a so that all points of the graph lying in that vertical strip, except possibly at $x = a$, lie inside the horizontal strip.



Why Do We Write $0 < |x - a|$?:

The condition

$$0 < |x - a|$$

means

$$x \neq a.$$

The value of the function at a itself does not determine the limit.

For instance,

$$f(x) = \begin{cases} x+2, & x \neq 1, \\ 100, & x = 1. \end{cases}$$

Then

$$f(1) = 100,$$

but

$$\boxed{\lim_{x \to 1} f(x) = 3}.$$

Thus the limit depends on nearby values of the function, not necessarily on the value at the point.

Understanding the Logical Order:

The statement

$$\boxed{\forall \varepsilon > 0, \quad \exists \delta > 0}$$

is extremely important.

It means that first an arbitrary positive number ε is chosen. Then we must be able to find a suitable positive number δ.

Usually,

$$\boxed{\delta = \delta(\varepsilon)},$$

that is, δ may depend on ε, but δ must not depend on x.

The logical structure is

$$\boxed{\text{Given } \varepsilon > 0, \text{ find } \delta > 0 \text{ such that } |f(x) - L| < \varepsilon \text{ whenever } 0 < |x - a| < \delta.}$$

The same ε-δ definition may be stated as follows:

Definition 2.2 (**Neighborhood Form of the Definition).** For every $\varepsilon > 0$, there is a $\delta > 0$ such that

$$f\left((a-\delta, a+\delta) \setminus \{a\}\right) \subseteq (L-\varepsilon, L+\varepsilon).$$

In words, every sufficiently small deleted neighborhood of a is mapped into the prescribed ε-neighborhood of L.

Examples Using ε-δ Definition

Problem 2.1. Prove that

$$\lim_{x\to 2}(3x+1) = 7.$$

Solution

We want to show that for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$0 < |x-2| < \delta$$

implies

$$|(3x+1)-7| < \varepsilon.$$

Now,

$$|(3x+1)-7| = |3x-6| = 3|x-2|.$$

We want

$$3|x-2| < \varepsilon.$$

This will happen if

$$|x-2| < \frac{\varepsilon}{3}.$$

Therefore choose

$$\boxed{\delta = \frac{\varepsilon}{3}}.$$

Then, if

$$0 < |x-2| < \delta,$$

we get

$$|(3x+1)-7| = 3|x-2| < 3\delta = 3\left(\frac{\varepsilon}{3}\right) = \varepsilon.$$

Hence,

$$\boxed{\lim_{x\to 2}(3x+1) = 7}.$$

Problem 2.2. Prove that

$$\lim_{x\to 3} x^2 = 9.$$

Solution

We have

$$|x^2-9| = |x-3||x+3|.$$

We want

$$|x-3||x+3| < \varepsilon.$$

Suppose we require

$$|x-3| < 1.$$

Then

$$2 < x < 4.$$

Therefore,

$$5 < x + 3 < 7,$$

so

$$|x + 3| < 7.$$

Thus,

$$|x^2 - 9| = |x - 3||x + 3| < 7|x - 3|.$$

To make this less than ε, require

$$|x - 3| < \frac{\varepsilon}{7}.$$

Therefore choose

$$\boxed{\delta = \min\left\{1, \frac{\varepsilon}{7}\right\}.}$$

Then

$$0 < |x - 3| < \delta$$

implies

$$|x^2 - 9| < 7|x - 3| < 7\delta \leq \varepsilon.$$

Hence,

$$\boxed{\lim_{x \to 3} x^2 = 9}.$$

Problem 2.3. Prove that

$$\lim_{x \to 1}(x^2 + x) = 2.$$

Solution

We have

$$|x^2 + x - 2| = |(x - 1)(x + 2)| = |x - 1||x + 2|.$$

Suppose

$$|x - 1| < 1.$$

Then

$$0 < x < 2.$$

Hence,

$$2 < x + 2 < 4,$$

so

$$|x + 2| < 4.$$

Therefore,

$$|x^2 + x - 2| < 4|x - 1|.$$

To make this less than ε, require

$$|x-1| < \frac{\varepsilon}{4}.$$

Hence choose

$$\boxed{\delta = \min\left\{1, \frac{\varepsilon}{4}\right\}.}$$

Then

$$0 < |x-1| < \delta$$

implies

$$|x^2 + x - 2| < 4|x-1| < 4\delta \leq \varepsilon.$$

Therefore,

$$\boxed{\lim_{x\to 1}(x^2+x) = 2}.$$

Problem 2.4. Prove that

$$\lim_{x\to a}(mx+c) = ma + c.$$

Solution

Consider

$$f(x) = mx + c.$$

We want to prove that

$$\lim_{x\to a} f(x) = ma + c.$$

Let

$$L = ma + c.$$

Then

$$\begin{aligned} |f(x) - L| &= |(mx+c) - (ma+c)| \\ &= |m(x-a)| \\ &= |m||x-a|. \end{aligned}$$

To have

$$|f(x) - L| < \varepsilon,$$

it is sufficient that

$$|x-a| < \frac{\varepsilon}{|m|}.$$

Thus, for $m \neq 0$, choose

$$\boxed{\delta = \frac{\varepsilon}{|m|}}.$$

Therefore,

$$\boxed{\lim_{x\to a}(mx+c) = ma + c.}$$

Uniqueness of Limit

Theorem 2.1. *If* $\lim_{x\to a} f(x)$ *exists, then it is unique.*

Proof

Suppose that $f(x)$ has two limits L and M as $x \to a$.
Thus, $\lim_{x\to a} f(x) = L$ and $\lim_{x\to a} f(x) = M$.
We shall prove that $L = M$.
If possible, assume that $L \neq M$. Then $|L - M| > 0$.
Choose $\varepsilon = \frac{1}{2}|L - M| > 0$.
Since $\lim_{x\to a} f(x) = L$, by the definition of limit, there exists a number $\delta_1 > 0$ such that $|f(x) - L| < \varepsilon$ whenever $0 < |x - a| < \delta_1$
Similarly, since $\lim_{x\to a} f(x) = M$, there exists a number $\delta_2 > 0$ such that $|f(x) - M| < \varepsilon$whenever $0 < |x - a| < \delta_2$
Let $\delta = \min\{\delta_1, \delta_2\}$.
Then, whenever $0 < |x - a| < \delta$, we have both $|f(x) - L| < \varepsilon$ and $|f(x) - M| < \varepsilon$.
Therefore, $|L-M| = |L-f(x)+f(x)-M| \leq |L-f(x)|+|f(x)-M|$ by the triangle inequality.
Hence, $|L - M| < \varepsilon + \varepsilon = 2\varepsilon$.
Using $\varepsilon = \frac{1}{2}|L - M|$, in the above, we obtain $|L - M| < 2\frac{1}{2}|L - M| = |L - M|$, which is not possible.
Therefore, our assumption $L \neq M$ is false. Hence,

$$\boxed{L = M}.$$

Thus, if the limit of a function exists at a point, it is unique. □

Bounded Function

Definition 2.3.

1. **Bounded above:** The function f is said to be *bounded above* on A if there exists a real number M such that

   $$\boxed{f(x) \leq M \text{ for every } x \in A}.$$

   Such a number M is called an *upper bound* of f on A.
2. **Bounded below:** The function f is said to be *bounded below* on A if there exists a real number m such that

   $$\boxed{m \leq f(x) \text{ for every } x \in A}.$$

   Such a number m is called a *lower bound* of f on A.
3. **Bounded:** The function f is said to be *bounded* on A if it is both bounded

above and bounded below. Thus, there exist real numbers m and M such that

$$\boxed{m \leq f(x) \leq M \text{ for every } x \in A}. \quad \textbf{(OR)}$$

Equivalently, there exists a real number $K > 0$ such that

$$\boxed{|f(x)| \leq K, \text{ that is, } -K \leq f(x) \leq K \text{ for every } x \in A}. \quad \textbf{(OR)}$$

Simply, a function is bounded if and only if its range is a bounded subset of $\mathbb{R}$.

Example 2.1. Consider $f(x) = \sin x$ on $\mathbb{R}$. Since $-1 \leq \sin x \leq 1$ for every $x \in \mathbb{R}$, we have $|\sin x| \leq 1$. Hence $f(x) = \sin x$ is bounded on $\mathbb{R}$.

Example 2.2. Consider $f(x) = \dfrac{1}{1+x^2}$ on $\mathbb{R}$. Since $x^2 \geq 0$, we have $1 + x^2 \geq 1$. Therefore, $0 < \dfrac{1}{1+x^2} \leq 1$, which implies $-1 < 0 < \dfrac{1}{1+x^2} \leq 1$, Hence $\left|\dfrac{1}{1+x^2}\right| \leq 1$, and so f is bounded on $\mathbb{R}$.

Example 2.3. Consider $f(x) = x$ on the interval $[-2, 2]$. Then $-2 \leq x \leq 2$, and hence $|f(x)| = |x| \leq 2$. Therefore $f(x) = x$ is bounded on $[-2, 2]$.
However, $f(x) = x$ is not bounded on $\mathbb{R}$, since its values can become arbitrarily large in both directions on $\mathbb{R}$.

Example 2.4. Consider $f(x) = \dfrac{1}{x}$ on the interval $(0, 1)$.
As x approaches 0, $\dfrac{1}{x}$ becomes arbitrarily large. Hence there is no $M > 0$ such that $\left|\dfrac{1}{x}\right| \leq M$ for every $x \in (0, 1)$.
Therefore, $f(x) = \dfrac{1}{x}$ is not bounded on $(0, 1)$.

Algebra of Limits

Lemma 2.1. *If $\lim\limits_{x \to a} f(x) = L$, then f is bounded in some deleted neighbourhood of a. Moreover, if $L \neq 0$, then $f(x) \neq 0$.*
That is, there exist constants $M > 0$ and $\delta > 0$ such that $|f(x)| \leq M$ whenever $0 < |x - a| < \delta$ and if $L \neq 0$, then $f(x) \neq 0$.

Proof

Since $\lim_{x\to a} f(x) = L$, for every $\varepsilon > 0$, there exists a $\delta > 0$ such that $|f(x) - L| < \varepsilon$ whenever $0 < |x - a| < \delta$.
Now, by using the fact that $||c| - |d|| \leq |c - d|$, we have

$$||f(x)| - |L|| \leq |f(x) - L| < \varepsilon.$$

Therefore,

$$||f(x)| - |L|| < \varepsilon \quad \text{or} \quad -\varepsilon \leq |f(x)| - |L| < \varepsilon \quad \text{or} \quad |L| - \varepsilon \leq |f(x)| < |L| + \varepsilon. \quad (1)$$

Thus from right side inequality of (1), we have

$$|f(x)| < |L| + \varepsilon \Longrightarrow f(x) \text{ is bounded, whenever } 0 < |x - a| < \delta.$$

Further if $L \neq 0$, take $\varepsilon = \dfrac{|L|}{2}$, therefore from (1), we have

$$|L| - \frac{|L|}{2} < |f(x)| < |L| + \frac{|L|}{2} \text{ or } \frac{|L|}{2} < |f(x)| < \frac{3|L|}{2}, \text{ which implies that} f(x) \neq 0.$$

□

Theorem 2.2. *If* $\lim_{x\to a} f(x) = L$ *and* $\lim_{x\to a} g(x) = M$, *then* $\lim_{x\to a}\Big(f(x) + g(x)\Big) = L + M$.
(KUD: Jan-2026, Feb-2025, Feb-2024, Mar-2022)(Question 1-QB)

Proof

Let $\varepsilon > 0$ be given.
Since $\lim_{x\to a} f(x) = L$, there exists $\delta_1 > 0$ such that $|f(x) - L| < \dfrac{\varepsilon}{2}$ whenever $0 < |x - a| < \delta_1$.
Since $\lim_{x\to a} g(x) = M$, there exists $\delta_2 > 0$ such that $|g(x) - M| < \dfrac{\varepsilon}{2}$ whenever $0 < |x - a| < \delta_2$.
Choose $\delta = \min\{\delta_1, \delta_2\}$.
Then, whenever $0 < |x - a| < \delta$, we have

$$\begin{aligned} |(f(x) + g(x)) - (L + M)| &= |(f(x) - L) + (g(x) - M)| \\ &\leq |f(x) - L| + |g(x) - M| \quad (\because\ |c + d| \leq |c| + |d|) \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

Hence,

$$\boxed{\lim_{x\to a}\Big(f(x) + g(x)\Big) = L + M}.$$

□

Theorem 2.3. *If* $\lim_{x\to a} f(x) = L$ *and* $\lim_{x\to a} g(x) = M$, *then* $\lim_{x\to a}\Big(f(x)-g(x)\Big) = L-M$.
(KUD: Feb-2023, Oct-2022)(Question 2-QB)

Proof

Let $\varepsilon > 0$ be given.
Since $\lim_{x\to a} f(x) = L$, there exists $\delta_1 > 0$ such that $|f(x) - L| < \frac{\varepsilon}{2}$ whenever $0 < |x-a| < \delta_1$.
Since $\lim_{x\to a} g(x) = M$, there exists $\delta_2 > 0$ such that $|g(x) - M| < \frac{\varepsilon}{2}$ whenever $0 < |x-a| < \delta_2$.
Choose $\delta = \min\{\delta_1, \delta_2\}$.
Then, whenever $0 < |x-a| < \delta$, we have

$$\begin{aligned}
|(f(x)-g(x))-(L-M)| &= |(f(x)-L)+(-g(x)+M)| \\
&\le |f(x)-L| + |-g(x)+M| \quad (\because\ |c+d| \le |c|+|d|) \\
&\le |f(x)-L| + |g(x)-M| \quad (\because\ |-c| = |c|) \\
&< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\
&= \varepsilon.
\end{aligned}$$

Therefore,

$$\boxed{\lim_{x\to a}\Big(f(x)-g(x)\Big) = L-M}.$$

□

Theorem 2.4. *If* $\lim_{x\to a} f(x) = L$ *and* $\lim_{x\to a} g(x) = M$, *then* $\lim_{x\to a} f(x)g(x) = LM$.
(Question 3-QB)

Proof

Let $\varepsilon > 0$ be given.
Since $\lim_{x\to a} f(x) = L$, there exists $\delta_1 > 0$ such that $|f(x) - L| < \frac{\varepsilon}{2(|M|+1)}$ whenever $0 < |x-a| < \delta_1$.
Since $\lim_{x\to a} g(x) = M$, there exists $\delta_2 > 0$ such that $|g(x) - M| < \frac{\varepsilon}{2(|L|+1)}$ whenever $0 < |x-a| < \delta_2$.
Also, since $\lim_{x\to a} f(x) = L$, there exists $\delta_3 > 0$ such that $|f(x)| < |L| + 1$ whenever $0 < |x-a| < \delta_3$. (see, Lemma 1 for $\varepsilon = 1$)
Choose $\delta = \min\{\delta_1, \delta_2, \delta_3\}$.

Then, whenever $0 < |x-a| < \delta$, we have

$$\begin{aligned}
|f(x)g(x) - LM| &= |f(x)g(x) - f(x)M + f(x)M - LM| \\
&= |f(x)(g(x)-M) + M(f(x)-L)| \\
&\leq |f(x)|\,|g(x)-M| + |M|\,|f(x)-L| \\
&< (|L|+1)\frac{\varepsilon}{2(|L|+1)} + |M|\frac{\varepsilon}{2(|M|+1)} \\
&< \frac{\varepsilon}{2} + 1\cdot\frac{\varepsilon}{2} \\
&= \varepsilon.
\end{aligned}$$

Hence,

$$\boxed{\lim_{x\to a} f(x)g(x) = LM}.$$

□

Theorem 2.5. *If* $\lim_{x\to a} f(x) = L$ *and* $\lim_{x\to a} g(x) = M$, *where* $M \neq 0$, *then*

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{L}{M}.$$

Proof

Let $\varepsilon > 0$ be given.

Since $\lim_{x\to a} f(x) = L$, there exists $\delta_1 > 0$ such that $|f(x) - L| < \dfrac{\varepsilon|M|}{4}$ whenever $0 < |x-a| < \delta_1$.

Since $\lim_{x\to a} g(x) = M$, there exists $\delta_2 > 0$ such that $|g(x) - M| < \dfrac{\varepsilon|M|^2}{4(|L|+1)}$ whenever $0 < |x-a| < \delta_2$.

Also, since $\lim_{x\to a} g(x) = M$ and $M \neq 0$, there exists $\delta_3 > 0$ such that $\dfrac{|M|}{2} < |g(x)|$ whenever $0 < |x-a| < \delta_3$. (see, Lemma 1)

Choose $\delta = \min\{\delta_1, \delta_2, \delta_3\}$.

Then, whenever $0 < |x-a| < \delta$, we have

$$\begin{aligned}
\left|\frac{f(x)}{g(x)} - \frac{L}{M}\right| &= \left|\frac{Mf(x) - Lg(x)}{Mg(x)}\right| = \left|\frac{Mf(x) - LM + LM - Lg(x)}{Mg(x)}\right| \\
&= \left|\frac{M(f(x)-L) + L(M-g(x))}{Mg(x)}\right| \\
&\leq \frac{|M|\,|f(x)-L| + |L|\,|g(x)-M|}{|M|\,|g(x)|}.
\end{aligned}$$

Since $|g(x)| > \frac{|M|}{2} \Longrightarrow \frac{1}{|g(x)|} < \frac{2}{|M|}$, we obtain

$$\begin{aligned}\left|\frac{f(x)}{g(x)} - \frac{L}{M}\right| &< \frac{2}{|M|^2}\left(|M|\,|f(x)-L| + |L|\,|g(x)-M|\right)\\ &< \frac{2}{|M|^2}\left(|M|\frac{\varepsilon|M|}{4} + |L|\frac{\varepsilon|M|^2}{4(|L|+1)}\right)\\ &= \frac{\varepsilon}{2} + \frac{|L|}{(|L|+1)}\frac{\varepsilon}{2} < \frac{\varepsilon}{2} + 1\cdot\frac{\varepsilon}{2}\\ &= \varepsilon.\end{aligned}$$

Therefore,

$$\boxed{\lim_{x\to a}\frac{f(x)}{g(x)} = \frac{L}{M}}.$$

□

Theorem 2.6. *If* $\lim_{x\to a} f(x) = L$, *then* $\lim_{x\to a}|f(x)| = |L|$.

Proof

Let $\varepsilon > 0$ be given.
Since $\lim_{x\to a} f(x) = L$, there exists a $\delta > 0$ such that $|f(x) - L| < \varepsilon$ whenever $0 < |x-a| < \delta$.
Therefore, whenever $0 < |x-a| < \delta$, we have $\big||f(x)| - |L|\big| \le |f(x) - L| < \varepsilon$.
Hence,

$$\boxed{\lim_{x\to a}|f(x)| = |L|}.$$

□

Note (Converse:)**.** The converse of the above theorem is *not true.*
That is, the statement $\lim_{x\to a}|f(x)| = |L|$ does not necessarily imply that $\lim_{x\to a} f(x) = L$.

Example 2.5 (Counterexample).

Define $f(x) = \begin{cases}1, & x > 0,\\ -1, & x < 0.\end{cases}$
We have $|f(x)| = 1$ for every $x \neq 0$. Therefore, $\lim_{x\to 0}|f(x)| = 1$.
However, $\lim_{x\to 0^+} f(x) = 1$ and $\lim_{x\to 0^-} f(x) = -1$.
Since, the right-hand and left-hand limits are different, $\lim_{x\to 0} f(x)$ does not exist.
Hence, the converse of the above theorem is false.
$\therefore$ $\boxed{\lim_{x\to a} f(x) = L \implies \lim_{x\to a}|f(x)| = |L|,}$ but $\boxed{\lim_{x\to a}|f(x)| = |L| \not\Longrightarrow \lim_{x\to a} f(x) = L.}$

Continuity

Intuitively, a function is continuous at a point if small changes in the input near that point produce small changes in the output. Thus, the values of $f(x)$ can be made arbitrarily close to $f(a)$ by taking x sufficiently close to a.

At a point a where the limit is applicable, continuity means

$$\boxed{\lim_{x\to a} f(x) = f(a).} \text{ That is, } \boxed{\lim_{x\to a^+} f(x) = \lim_{x\to a^-} f(x) = f(a).}$$

In particular, $f(a)$ must be defined, the limit must exist, and the limit must equal $f(a)$.

Definition 2.4 (ε-δ **Definition of Continuity).** Let $f : D \to \mathbb{R}$ and let $a \in D$. We say that f is **continuous at** a if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that $|f(x) - f(a)| < \varepsilon$ whenever $x \in D$ and $|x - a| < \delta$.
Symbolically, $\boxed{\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in D, \quad |x - a| < \delta \implies |f(x) - f(a)| < \varepsilon.}$

(Question 1-QB)

Remark. The conditions $|x - a| < \delta$ and $0 < |x - a| < \delta$ give equivalent definitions of continuity at a, provided that $f(a)$ is defined.
Indeed, if $|x - a| < \delta \implies |f(x) - f(a)| < \varepsilon$, then the same implication clearly holds when $0 < |x - a| < \delta$.
Conversely, suppose that $0 < |x-a| < \delta \implies |f(x)-f(a)| < \varepsilon$. For $|x-a| < \delta$, there are two cases:

- If $x \neq a$, then $0 < |x - a| < \delta$, so $|f(x) - f(a)| < \varepsilon$ by assumption.
- If $x = a$, then $|f(x) - f(a)| = |f(a) - f(a)| = 0 < \varepsilon$.

Hence the inequality holds whenever $|x-a| < \delta$. Thus, including or excluding $x = a$ does not change the definition of continuity.

A function is **continuous on** D if it is continuous at every point of D.
Continuity on a closed interval: A function $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$ if it is continuous at every point of (a, b), right-continuous at a, and left-continuous at b. Thus, at the endpoints, we require $\lim_{x\to a^+} f(x) = f(a)$ and $\lim_{x\to b^-} f(x) = f(b)$.

Example 2.6. Find the indicated constant so that the function is continuous at the given point:

1. $f(x) = \begin{cases} \lambda x^2, & x \leq 2, \\ 3, & x > 2; \end{cases}$ $x = 2$; find λ.
2. $f(x) = \begin{cases} 3x - 2, & x \leq 1, \\ k, & x > 1; \end{cases}$ $x = 1$; find k.
3. $f(x) = \begin{cases} \dfrac{\sin 3x}{x}, & x \neq 0, \\ K, & x = 0; \end{cases}$ $x = 0$; find K.

(Question 5-QB)

1.

Solution

For continuity at $x = 2$, we require

$$\lim_{x\to 2^-} f(x) = \lim_{x\to 2^+} f(x) = f(2).$$

Here,

$$\text{L.H.L.} = \lim_{x\to 2^-} \lambda x^2 = 4\lambda, \qquad \text{R.H.L.} = \lim_{x\to 2^+} 3 = 3,$$

and $f(2) = 4\lambda$.
Therefore, f is continuous at 2 if

$$4\lambda = 3 \implies \boxed{\lambda = \frac{3}{4}.}$$

For this value, L.H.L. = R.H.L. = $f(2) = 3$.

2.

Solution

For continuity at $x = 1$, we require

$$\lim_{x\to 1^-} f(x) = \lim_{x\to 1^+} f(x) = f(1).$$

Here,

$$\text{L.H.L.} = \lim_{x\to 1^-} (3x - 2) = 1, \qquad \text{R.H.L.} = \lim_{x\to 1^+} k = k,$$

and $f(1) = 3(1) - 2 = 1$.
Therefore, f is continuous at 1 if

$$\boxed{k = 1.}$$

For this value, L.H.L. = R.H.L. = $f(1) = 1$.

3.

Solution

For continuity at $x = 0$, we require

$$\lim_{x\to 0^-} f(x) = \lim_{x\to 0^+} f(x) = f(0).$$

Using the standard limit

$$\lim_{t\to 0} \frac{\sin t}{t} = 1 \qquad \text{(angles in radians)},$$

we obtain

$$\text{L.H.L.} = \lim_{x\to 0^-} \frac{\sin 3x}{x} = 3 \lim_{x\to 0^-} \frac{\sin 3x}{3x} = 3,$$

and

$$\text{R.H.L.} = \lim_{x\to 0^+} \frac{\sin 3x}{x} = 3 \lim_{x\to 0^+} \frac{\sin 3x}{3x} = 3.$$

Since $f(0) = K$, continuity at 0 requires

$$\boxed{K = 3.}$$

For this value, L.H.L. = R.H.L. = $f(0) = 3$.

Theorem 2.7 (Subdivision Theorem)**.** *Let $f : [a, b] \to \mathbb{R}$ be continuous. Then, for every $\varepsilon > 0$, there exists a finite partition $a = a_0 < a_1 < \cdots < a_m = b$ such that $|f(u) - f(v)| < \varepsilon$ whenever u and v belong to the same subinterval $[a_{j-1}, a_j]$.*

Proof

Fix $\varepsilon > 0$. Call a finite partition *suitable* if $|f(u) - f(v)| < \varepsilon$ whenever u and v belong to the same subinterval.
Suppose, for a contradiction, that $[a, b]$ has no suitable finite partition. Bisect $[a, b]$ into $\left[a, \frac{a+b}{2}\right]$ and $\left[\frac{a+b}{2}, b\right]$. If both halves had suitable finite partitions, combining their partition points would give a suitable finite partition of $[a, b]$. Therefore, at least one half has no suitable finite partition. Choose such a half and denote it by I_1. Then $|I_1| = \dfrac{b-a}{2}$, where $|I_1|$ denotes the length of I_1.
Now bisect I_1. By the same reasoning, at least one of its halves has no suitable finite partition. Choose such a half and denote it by I_2. Continuing this process, we obtain nested closed intervals

$$[a, b] \supseteq I_1 \supseteq I_2 \supseteq \cdots \supseteq I_n \supseteq \cdots$$

such that each I_n has no suitable finite partition and $|I_n| = \dfrac{b-a}{2^n}$.
By the nested interval property, there exists a point c belonging to every I_n. In particular, $c \in [a, b]$. Since f is continuous at c, there exists $\delta > 0$ such that

$$x \in [a, b] \text{ and } |x - c| < \delta \quad \Longrightarrow \quad |f(x) - f(c)| < \frac{\varepsilon}{2}.$$

Choose n sufficiently large that

$$|I_n| = \frac{b-a}{2^n} < \delta.$$

Since c and any point $x \in I_n$ lie in the same interval, their distance cannot exceed its length. Thus

$$|x - c| \leq |I_n| < \delta.$$

Consequently,

$$|f(x) - f(c)| < \frac{\varepsilon}{2} \qquad \text{for every } x \in I_n.$$

For any $u, v \in I_n$, the triangle inequality now gives

$$\begin{aligned} |f(u) - f(v)| &\leq |f(u) - f(c)| + |f(c) - f(v)| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

This contradicts the choice of I_n as an interval having no suitable finite partition. Hence $[a, b]$ has the required finite partition. □

Theorem 2.8. *Every continuous function defined on $[a, b]$ is bounded on $[a, b]$.*
(KUD: Feb-2025)(Question 6-QB)

Proof

Let $f : [a, b] \to \mathbb{R}$ be continuous.
Let $\varepsilon > 0$ be given. Then there exists a finite partition $a = a_0 < a_1 < \cdots < a_n = b$ such that $|f(u) - f(v)| < \varepsilon$ whenever u and v belong to the same subinterval.
Let $x \in [a, b]$. Then $x \in [a_{k-1}, a_k]$ for some $k \in \{1, \ldots, n\}$. We can write

$$\begin{aligned} f(x) &= f(x) + f(a_0) \underbrace{-f(a_0) + f(a_1)} \overbrace{-f(a_1) + f(a_2)} \underbrace{-f(a_2) + \cdots} \\ &\quad \overbrace{\cdots + f(a_{k-1})} \underbrace{-f(a_{k-1}) + f(a_k)} - f(a_k) \\ &= f(a_0) + \sum_{j=1}^{k} \Big(f(a_j) - f(a_{j-1}) \Big) + \Big(f(x) - f(a_k) \Big). \end{aligned}$$

By the triangle inequality, we have

$$\begin{aligned} |f(x)| &\le |f(a_0)| + \sum_{j=1}^{k} |f(a_j) - f(a_{j-1})| + |f(x) - f(a_k)| \\ &< |f(a_0)| + k\varepsilon + \varepsilon \\ &= |f(a)| + (k+1)\varepsilon \\ &\le |f(a)| + (n+1)\varepsilon, \text{ which is independent of } x. \end{aligned}$$

Thus, setting $M = |f(a)| + (n+1)\varepsilon$, we obtain $|f(x)| \le M$ for every $x \in [a, b]$. Therefore, f is bounded on $[a, b]$. □

Definition 2.5 (Supremum and Infimum of a Function). Let f be a real-valued function defined on a nonempty set A.

1. **Supremum.** If f is bounded above on A, the **supremum** of f on A is the least upper bound of its range, denoted by $\sup_{x \in A} f(x)$ or, simply $\sup f(x)$.
   Thus, $M = \sup_{x \in A} f(x)$ if:
   (a) $f(x) \le M$ for every $x \in A$;
   (b) for every $\varepsilon > 0$, there exists $x \in A$ such that $M - \varepsilon < f(x) \le M$.

☞ : If $f(c) = M$ for some $c \in A$, the supremum is **attained**, and M is called the **maximum value** of f on A.

2. **Infimum.** If f is bounded below on A, the **infimum** of f on A is the greatest lower bound of its range, denoted by $\inf_{x \in A} f(x)$. or, simply $\inf f(x)$.

Thus, $m = \inf_{x\in A} f(x)$ if:
(a) $m \le f(x)$ for every $x \in A$;
(b) for every $\varepsilon > 0$, there exists $x \in A$ such that $m \le f(x) < m + \varepsilon$.

☞ : If $f(d) = m$ for some $d \in A$, the infimum is **attained**, and m is called the **minimum value** of f on A.

Example 2.7 (Supremum and Infimum)**.**

1. **Both supremum and infimum are attained.**
   Let $f(x) = x^2$ on $[0, 2]$. Its range is $[0, 4]$. Therefore,
   $$\sup_{x\in[0,2]} f(x) = 4 \quad \text{and} \quad \inf_{x\in[0,2]} f(x) = 0.$$
   Since $f(2) = 4$ and $f(0) = 0$, the supremum is attained at $x = 2$, and the infimum is attained at $x = 0$. Thus, the maximum and minimum values of $f(x)$ on $[0, 2]$ are 4 and 0, respectively.
2. **Neither supremum nor infimum is attained.**
   Let $f(x) = x^2$ on $(0, 2)$. Its range is $(0, 4)$. Therefore,
   $$\sup_{x\in(0,2)} f(x) = 4 \quad \text{and} \quad \inf_{x\in(0,2)} f(x) = 0.$$
   The values of $f(x)$ approach 4 as $x \to 2^-$ and 0 as $x \to 0^+$, but
   $$0 < f(x) < 4 \qquad \text{for every } x \in (0, 2).$$
   Hence, neither bound is attained, and f has neither a maximum nor a minimum value on $(0, 2)$.
3. **Supremum is attained, but infimum is not attained.**
   Let $f(x) = 1 - x^2$ on $(-1, 1)$. Its range is $(0, 1]$. Therefore,
   $$\sup_{x\in(-1,1)} f(x) = 1 \quad \text{and} \quad \inf_{x\in(-1,1)} f(x) = 0.$$
   Since $f(0) = 1$, the supremum is attained at $x = 0$. However, $f(x) > 0$ for every $x \in (-1, 1)$, and $f(x) \to 0$ as $x \to 1^-$. Thus, the maximum value is 1, but there is no minimum value.
4. **Infimum is attained, but supremum is not attained.**
   Let $f(x) = \dfrac{x}{1+x}$ on $[0, \infty)$. Its range is $[0, 1)$. Therefore,
   $$\sup_{x\in[0,\infty)} f(x) = 1 \quad \text{and} \quad \inf_{x\in[0,\infty)} f(x) = 0.$$
   Since $f(0) = 0$, the infimum is attained at $x = 0$. However, $f(x) < 1$ for every $x \ge 0$, and $f(x) \to 1$ as $x \to \infty$. Thus, the minimum value is 0, but there is no maximum value of $f(x)$ on $[0, \infty)$.

Remark. The supremum and infimum need not belong to the range of f. When attained, they are the maximum and minimum values of f on given domain of f, respectively.

Theorem 2.9 (Extreme Value Theorem)**.** *Prove that a function continuous on $[a,b]$ attains its bounds.* (**OR**)
Prove that a function continuous on $[a,b]$ attains its supremum and infimum at least once in $[a,b]$.

(KUD: Jan-2026)(Question 8-QB)

Proof

Since f is continuous on $[a,b]$, the boundedness theorem implies that f is bounded on $[a,b]$. Hence its supremum and infimum exist as real numbers. Let on $[a,b]$

$$M = \sup f(x) \quad \text{and} \quad m = \inf f(x).$$

Suppose, for a contradiction, that f does not attain M.
Since $f(x) \leq M$, we then have $f(x) < M$ for every $x \in [a,b]$.
Thus $M - f(x) > 0$ for every $x \in [a,b]$, so the function $g(x) = \dfrac{1}{M - f(x)}$ is continuous on $[a,b]$.
By the boundedness theorem, g is bounded. Therefore, there exists a real number $K > 0$ such that $\dfrac{1}{M - f(x)} \leq K$ for every $x \in [a,b]$.
It follows that $M - f(x) \geq \dfrac{1}{K}$, and hence $f(x) \leq M - \dfrac{1}{K} < M$ for every $x \in [a,b]$.
Thus $M - \frac{1}{K}$ is an upper bound for f on $[a,b]$ smaller than M, contradicting the fact that M is the least upper bound.
Therefore, there exists $\alpha \in [a,b]$ such that $f(\alpha) = M$.
$\Longrightarrow$ f attains supremum at least once on $[a,b]$.
Similarly, one can prove that f attains infimum at least once on $[a,b]$. □

Theorem 2.10 (Intermediate Value Theorem)**.** *Let $f : [a,b] \to \mathbb{R}$ be continuous, with $f(a) \neq f(b)$. Then f takes every value between $f(a)$ and $f(b)$ at least once in (a,b).* (**OR**)
Let $f : [a,b] \to \mathbb{R}$ be continuous, with $f(a) \neq f(b)$. Then for every real number k between $f(a)$ and $f(b)$, there exists $c \in (a,b)$ such that $f(c) = k$.

(KUD: Feb-2025)(Question 7-QB)

Proof

Suppose first that $f(a) < f(b)$, and let k be any number such that $f(a) < k < f(b)$.
Define $g(x) = f(x) - k$ on $[a,b]$.
Since f is continuous and k is a constant, g is continuous on $[a,b]$.

Moreover, $g(a) = f(a) - k < 0$ and $g(b) = f(b) - k > 0$.
Thus $g(a)$ and $g(b)$ have opposite signs.
$\Longrightarrow$ there exists $c \in (a, b)$ such that $g(c) = 0$.
Therefore, $g(c) = f(c) - k = 0 \quad \Longrightarrow \quad f(c) = k$.
Again, if $f(a) > f(b)$, take k such that $f(b) < k < f(a)$. Then $g(a) > 0$ and $g(b) < 0$, so again there exists $c \in (a, b)$ with $f(c) = k$.
Hence f takes every value between $f(a)$ and $f(b)$ at least once in (a, b). □

Problem 2.5. Show that $f(x) = x^2$ takes the value 5 on $[1, 3]$ using the Intermediate Value Theorem.

(Question 9(i)-QB)

Solution

The function $f(x) = x^2$ is a polynomial, so it is continuous on $[1, 3]$.
Also, $f(1) = (1)^2 = 1$ and $f(3) = (3)^2 = 9$.
Since $f(1) = 1 < 5 < 9 = f(3)$, the Intermediate Value Theorem guarantees that there exists $c \in (1, 3)$ such that $f(c) = 5$.
To find c, we solve $f(c) = c^2 = 5 \quad \Longrightarrow \quad c = \pm\sqrt{5}$.
Since $c \in (1, 3)$, we obtain $\boxed{c = \sqrt{5}.}$
Thus, $f(x) = x^2$ takes the value 5 at $x = \sqrt{5} \in [1, 3]$.

Problem 2.6. Show that $f(x) = \sin x$ takes the value 0.5 on $\left[0, \frac{\pi}{2}\right]$ using the Intermediate Value Theorem.

(Question 9(iv)-QB)

Solution

The function $f(x) = \sin x$ is continuous on $\left[0, \frac{\pi}{2}\right]$.
Also, $f(0) = \sin 0 = 0$ and $f\left(\frac{\pi}{2}\right) = \sin\frac{\pi}{2} = 1$.
Since $f(0) = 0 < 0.5 < 1 = f\left(\frac{\pi}{2}\right)$, the Intermediate Value Theorem guarantees that there exists $c \in \left(0, \frac{\pi}{2}\right)$ such that $f(c) = 0.5$.
To find c, we solve $f(c) = \sin c = \frac{1}{2} \Longrightarrow \boxed{c = \frac{\pi}{6}.}$.
Thus, $f(x) = \sin x$ takes the value 0.5 at $x = \frac{\pi}{6} \in \left[0, \frac{\pi}{2}\right]$.

Problem 2.7. Show that $x^2 - 2x - 5 = 0$ has at least one root in $(3, 4)$ using the Intermediate Value Theorem.

(Question 10(i)-QB)

Solution

Let $f(x) = x^2 - 2x - 5$. Since f is a polynomial, it is continuous on $[3, 4]$.
Also, $f(3) = 9 - 6 - 5 = -2$ and $f(4) = 16 - 8 - 5 = 3$.
Since $f(3) = -2 < 0 < 3 = f(4)$, the Intermediate Value Theorem guarantees that there exists $c \in (3, 4)$ such that $f(c) = 0$. Hence, $x^2 - 2x - 5 = 0$ has at least one root in $(3, 4)$.
To find c, we solve $c^2 - 2c - 5 = 0$ using the quadratic formula. Therefore,

$$c = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(-5)}}{2(1)} = \frac{2 \pm \sqrt{4 + 20}}{2}$$
$$= \frac{2 \pm \sqrt{24}}{2} = \frac{2 \pm 2\sqrt{4 \times 6}}{2} = \frac{2 \pm 2\sqrt{6}}{2} = 1 \pm \sqrt{6}$$

Since $c \in (3, 4)$, we obtain $\boxed{c = 1 + \sqrt{6}.}$
Thus, $x^2 - 2x - 5 = 0$ has a root $x = 1 + \sqrt{6} \in (3, 4)$.

Derivative of a Function

The derivative measures the instantaneous rate of change of a function. Geometrically, it gives the slope of the tangent to the graph at a point, when the derivative exists.

The terms *derivable* and *differentiable* have the same meaning.

Definition 2.6 (Derivative at a Point). Let f be defined in an open interval containing a. The function f is said to be **differentiable at** a if the limit

$$\boxed{f'(a) = \lim_{h \to 0} \frac{f(a + h) - f(a)}{h}}$$

exists as a finite real number.
This limit is called the **derivative of** f **at** a. Equivalently,

$$\boxed{f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}.}$$

The expression $\dfrac{f(a + h) - f(a)}{h}$ is called the **difference quotient**. It represents the slope of the secant line joining $P = (a, f(a))$ and $Q = (a + h, f(a + h))$. If the difference quotient approaches a finite value as $h \to 0$, that value is the slope of the tangent at P. The tangent line then has equation $\boxed{y - f(a) = f'(a)(x - a)}$.

Left-hand and Right-hand Derivatives

Definition 2.7 (**Left-hand Derivative**). The **left-hand derivative** of f at a is

$$\boxed{Lf'(a) = \lim_{x \to a^-} \frac{f(x) - f(a)}{x - a},}$$

provided the limit exists as a finite real number.
Putting $x = a - h$, where $h > 0$, we obtain

$$\boxed{Lf'(a) = \lim_{h \to 0^+} \frac{f(a-h) - f(a)}{-h}.}$$

Definition 2.8 (**Right-hand Derivative**). The **right-hand derivative** of f at a is

$$\boxed{Rf'(a) = \lim_{x \to a^+} \frac{f(x) - f(a)}{x - a},}$$

provided the limit exists as a finite real number.
Putting $x = a + h$, where $h > 0$, we obtain

$$\boxed{Rf'(a) = \lim_{h \to 0^+} \frac{f(a+h) - f(a)}{h}.}$$

Remark (Criterion for Differentiability)**.** A function is differentiable at an interior point a if and only if both one-sided derivatives exist as finite real numbers and are equal: $\boxed{Lf'(a) = Rf'(a).}$ Their common value is $f'(a)$.
If the one-sided derivatives are unequal, or either fails to exist as a finite real number, then f is not differentiable at a.

Derivability in an Interval

Definition 2.9 (**Derivability on an Open Interval**). A function f is said to be **derivable on** (a, b) if it is differentiable at every point of (a, b).

Definition 2.10 (**Derivability on a Closed Interval**). A function f is said to be **derivable on** $[a, b]$ if:

(i) f is differentiable at every point of (a, b);
(ii) the right-hand derivative $Rf'(a)$ exists as a finite real number;
(iii) the left-hand derivative $Lf'(b)$ exists as a finite real number.

☞ Similarly, derivability on $[a, b)$ requires differentiability on (a, b) and a finite right-hand derivative at a. Derivability on $(a, b]$ requires differentiability on (a, b) and a finite left-hand derivative at b.

Differentiability and Continuity

Theorem 2.11. *If a function is differentiable at a point, then it is continuous at that point.*

(Question 4-QB)

Proof

Suppose that f is differentiable at a. Then $f'(a) = \lim_{x\to a} \dfrac{f(x)-f(a)}{x-a}$ exists as a finite real number.
For $x \neq a$, we have

$$f(x) - f(a) = \frac{f(x)-f(a)}{x-a}\,(x-a).$$

Taking limits as $x \to a$, we obtain

$$\lim_{x\to a}(f(x)-f(a)) = \left(\lim_{x\to a}\frac{f(x)-f(a)}{x-a}\right)\left(\lim_{x\to a}(x-a)\right)$$
$$\lim_{x\to a} f(x) - \lim_{x\to a} f(a) = f'(a)\cdot 0$$
$$\lim_{x\to a} f(x) - f(a) = 0.$$

Therefore, $\lim_{x\to a} f(x) = f(a)$. Hence, f is continuous at a. □

Note (Converse:)**.** The converse of the above theorem is *not true*.
That is, continuity at a point does not necessarily imply that differentiability at that point.

Sharp Corners: Four Simple Counterexamples

At a **sharp corner**, the graph is continuous but the left-hand and right-hand derivatives are finite and unequal. Thus, the two sides give different limiting tangent slopes.

Example 2.8 (Counterexample 1). Show that $f(x) = |x|$ is continuous at 0 but not differentiable there.

Solution

We have

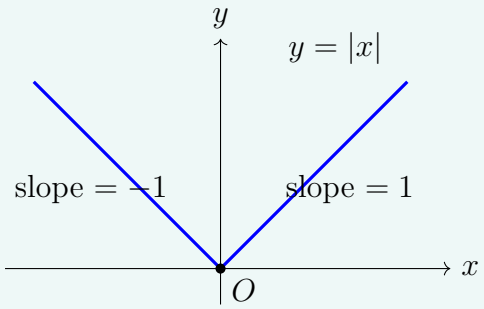
$$f(x) = \begin{cases} -x, & x < 0, \\ x, & x \geq 0, \end{cases} \text{ and } f(0) = 0.$$

Continuity at 0:
Taking $h > 0$,

L.H.L. $= \lim_{h\to 0^+} f(0-h) = \lim_{h\to 0^+} h = 0$, and R.H.L. $= \lim_{h\to 0^+} f(0+h) = \lim_{h\to 0^+} h = 0$.
Thus,

$$\text{L.H.L.} = \text{R.H.L.} = f(0) = 0.$$

Hence, f is continuous at 0.
Differentiability at 0:
The left-hand derivative is $Lf'(0) = \lim_{h\to 0^+} \dfrac{f(-h)-f(0)}{-h} = \lim_{h\to 0^+} \dfrac{h}{-h} = -1.$
The right-hand derivative is $Rf'(0) = \lim_{h\to 0^+} \dfrac{f(h)-f(0)}{h} = \lim_{h\to 0^+} \dfrac{h}{h} = 1.$
Since $Lf'(0) \neq Rf'(0)$, f is not differentiable at 0.
Geometrical interpretation. The left branch has slope -1, while the right branch has slope 1. They meet at a sharp corner at $(0,0)$, so there is no single tangent slope there. Hence, f is not differentiable at $x = 0$.



Example 2.9 (Counterexample 2). Show that $f(x) = |x-1|$ is continuous at 1 but not differentiable there.

Solution

We have

$$f(x) = \begin{cases} 1-x, & x < 1, \\ x-1, & x \geq 1, \end{cases} \text{ and } f(1) = 0.$$

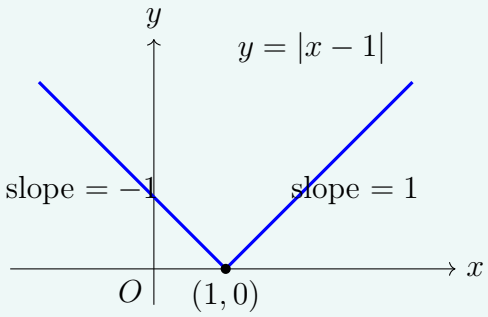
Continuity at 1:
For $h > 0$, $f(1-h) = |-h| = h$, and $f(1+h) = |h| = h$.
Therefore, L.H.L. $= \lim_{h\to 0^+} f(1-h) = 0$, and R.H.L. $= \lim_{h\to 0^+} f(1+h) = 0$.
Thus, L.H.L. $=$ R.H.L. $= f(1) = 0$, so f is continuous at 1.
Differentiability at 1:

$$Lf'(1) = \lim_{h\to 0^+} \frac{f(1-h)-f(1)}{-h} = \lim_{h\to 0^+} \frac{h}{-h} = -1,$$

whereas

$$Rf'(1) = \lim_{h\to 0^+} \frac{f(1+h)-f(1)}{h} = \lim_{h\to 0^+} \frac{h}{h} = 1.$$

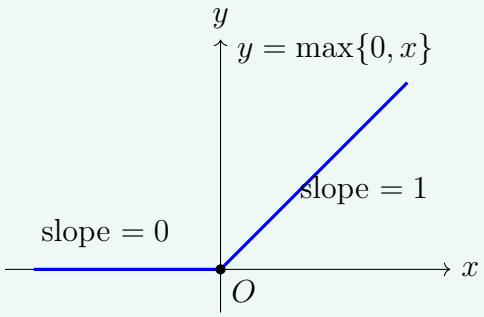
Since the one-sided derivatives are unequal, f is not differentiable at 1.
Geometrical interpretation. The graph is the graph of $y = |x|$ shifted one unit to the right. Its sharp corner is at $(1,0)$. The two branch slopes are -1 and 1. Hence, f is not differentiable at $x = 1$.



Example 2.10 (Counterexample 3). Show that $f(x) = \begin{cases} 0, & x < 0, \\ x, & x \geq 0 \end{cases}$ is continuous at 0 but not differentiable there.

Solution

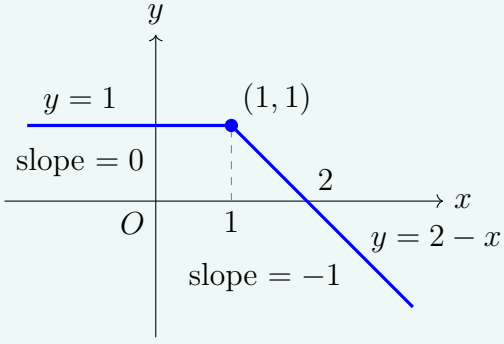
Geometrical interpretation. The left branch is horizontal and has slope 0. The right branch is $y = x$ and has slope 1. Hence, f is not differentiable at $x = 0$.



Example 2.11 (Counterexample 4). Show that $f(x) = \begin{cases} 1, & x < 1, \\ 2 - x, & x \geq 1 \end{cases}$ is continuous at $x = 1$ but not differentiable there.

Solution

Geometrical interpretation. The left branch is the horizontal line $y = 1$, with slope 0. The right branch is the line $y = 2 - x$, with slope -1. Hence, f is not differentiable at $x = 1$.



Remark. The examples show that

$$\boxed{\text{Differentiability implies continuity, but continuity need not imply differentiability.}}$$

A sharp corner is one reason for failure of differentiability. Other reasons include discontinuity, an infinite limiting slope, or an oscillating difference quotient.

Methods for Proving Non-differentiability at a Point

☞ To show that a function f is not differentiable at $x = a$, we may use either of the following methods:

1. **Show that f is not continuous at a.**

   Find $\lim\limits_{x\to a^-} f(x)$, $\lim\limits_{x\to a^+} f(x)$ and $f(a)$.

   If these three values do not exist and agree, then f is not continuous at a.

   Contrapositive Statement.

   $$\boxed{f \text{ is not continuous at } a \implies f \text{ is not differentiable at } a.}$$

   This is the contrapositive of the theorem:

   $$\boxed{f \text{ is differentiable at } a \implies f \text{ is continuous at } a.}$$

   A statement and its contrapositive are logically equivalent.

2. **Examine the left-hand and right-hand derivatives.**

   Calculate $Lf'(a) = \lim\limits_{h\to 0^+} \dfrac{f(a-h)-f(a)}{-h}$ and $Rf'(a) = \lim\limits_{h\to 0^+} \dfrac{f(a+h)-f(a)}{h}$.

   If these limits exist as finite real numbers but $Lf'(a) \neq Rf'(a)$, then f is not differentiable at a. The same conclusion holds if either limit fails to exist as a finite real number.

Problem 2.8. If $f(x) = \begin{cases} \dfrac{x(e^{1/x}-e^{-1/x})}{e^{1/x}+e^{-1/x}}, & x \neq 0 \\ 10, & x = 0 \end{cases}$, then show that f is not differentiable at $x = 0$.

(Question 5(i)-QB)

Solution

Continuity at $x = 0$:

Putting $x = 0 - h$, we obtain

$$\begin{aligned}\text{L.H.L.} &= \lim_{x\to 0^-} f(x)\\ &= \lim_{h\to 0^+}(-h)\frac{e^{-1/h}-e^{1/h}}{e^{-1/h}+e^{1/h}}\\ &= \lim_{h\to 0^+}(-h)\frac{\cancel{e^{1/h}}(e^{-2/h}-1)}{\cancel{e^{1/h}}(e^{-2/h}+1)} \quad (\because\ e^{-2/h}\to 0 \text{ as } h\to 0^+)\\ &= 0\cdot\frac{0-1}{0+1} = 0.\end{aligned}$$

Putting $x = 0 + h$, we obtain

$$\begin{aligned}\text{R.H.L.} &= \lim_{x\to 0^+} f(x)\\ &= \lim_{h\to 0^+} h\frac{e^{1/h}-e^{-1/h}}{e^{1/h}+e^{-1/h}}\\ &= \lim_{h\to 0^+} h\frac{\cancel{e^{1/h}}(1-e^{-2/h})}{\cancel{e^{1/h}}(1+e^{-2/h})} \quad (\because\ e^{-2/h}\to 0 \text{ as } h\to 0^+)\\ &= 0\cdot\frac{1-0}{1+0} = 0.\end{aligned}$$

But $f(0) = 10$. Thus,

$$\text{L.H.L.} = \text{R.H.L.} = 0 \neq f(0).$$

Hence, $\boxed{f \text{ is not continuous at } x = 0, \text{ and therefore it is not differentiable there.}}$

Problem 2.9. If $f(x) = \begin{cases}\dfrac{x(e^{1/x}-1)}{e^{1/x}+1}, & x \neq 0\\ 0, & x = 0\end{cases}$, then show that f is not differentiable at $x = 0$.

(Question 5(ii)-QB)

Solution

Differentiability at $x = 0$:
Putting $x = 0 - h$, we obtain the left-hand derivative

$$\begin{aligned}\text{L.H.D.} &= \lim_{h\to 0^+}\frac{f(-h)-f(0)}{-h}\\ &= \lim_{h\to 0^+}\frac{(-h)\dfrac{e^{-1/h}-1}{e^{-1/h}+1}-0}{-h}\\ &= \lim_{h\to 0^+}\frac{e^{-1/h}-1}{e^{-1/h}+1} \quad (\because\ e^{-1/h}\to 0 \text{ as } h\to 0^+)\\ &= \frac{0-1}{0+1} = -1.\end{aligned}$$

Putting $x = 0 + h$, we obtain the right-hand derivative

$$\begin{aligned}
\text{R.H.D.} &= \lim_{h\to 0^+} \frac{f(h) - f(0)}{h} \\
&= \lim_{h\to 0^+} \frac{h\dfrac{e^{1/h} - 1}{e^{1/h} + 1} - 0}{h} \\
&= \lim_{h\to 0^+} \frac{e^{1/h} - 1}{e^{1/h} + 1} \\
&= \lim_{h\to 0^+} \frac{\cancel{e^{1/h}}(1 - e^{-1/h})}{\cancel{e^{1/h}}(1 + e^{-1/h})} \quad (\because\ e^{-1/h} \to 0 \text{ as } h \to 0^+) \\
&= \frac{1 - 0}{1 + 0} = 1.
\end{aligned}$$

Since L.H.D. $\neq$ R.H.D., $\boxed{f \text{ is not differentiable at } x = 0.}$

Problem 2.10. Show that $f(x) = \begin{cases} x\cos(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases}$ is continuous at $x = 0$ but not differentiable there.

(Question 5(iii)-QB)

Solution

Continuity at $x = 0$:
Putting $x = 0 - h$, we obtain

$$\begin{aligned}
\text{L.H.L.} &= \lim_{x\to 0^-} f(x) \\
&= \lim_{h\to 0^+} (-h)\cos(-1/h) \\
&= \lim_{h\to 0^+} -h\cos(1/h) \quad (\because\ \cos(-t) = \cos t) \\
&= 0 \times \text{ a finite value keeps oscillating between } -1 \text{ and } 1 \\
&= 0.
\end{aligned}$$

Putting $x = 0 + h$, we obtain

$$\begin{aligned}
\text{R.H.L.} &= \lim_{x\to 0^+} f(x) \\
&= \lim_{h\to 0^+} h\cos(1/h) = 0 \\
&= 0 \times \text{ a finite value keeps oscillating between } -1 \text{ and } 1 \\
&= 0.
\end{aligned}$$

Since $f(0) = 0$,

$$\text{L.H.L.} = \text{R.H.L.} = f(0) = 0.$$

Therefore, f is continuous at $x = 0$.
Differentiability at $x = 0$:

Putting $x = 0 - h$, we obtain the left-hand derivative

$$\begin{aligned}\text{L.H.D.} &= \lim_{h\to 0^+} \frac{f(-h) - f(0)}{-h} \\ &= \lim_{h\to 0^+} \frac{-h\cos(-1/h) - 0}{-h} \\ &= \lim_{h\to 0^+} \cos(1/h) \\ &= \text{ a finite value keeps oscillating between } -1 \text{ and } 1.\end{aligned}$$

Putting $x = 0 + h$, we obtain the right-hand derivative

$$\begin{aligned}\text{R.H.D.} &= \lim_{h\to 0^+} \frac{f(h) - f(0)}{h} \\ &= \lim_{h\to 0^+} \frac{h\cos(1/h) - 0}{h} \\ &= \lim_{h\to 0^+} \cos(1/h) \\ &= \text{ a finite value keeps oscillating between } -1 \text{ and } 1.\end{aligned}$$

Thus, in both case L.H.D. and R.H.D. do not approach a single value.
Hence, neither the left-hand nor the right-hand derivative exists, and

$$\therefore \boxed{f \text{ is continuous at 0 but not differentiable there.}}$$

Problem 2.11. Prove that $f(x) = \begin{cases} x^2\sin(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases}$ is continuous and differentiable at $x = 0$.

(Question 5(iv)-QB)

Solution

Differentiability at $x = 0$:
Putting $x = 0 - h$, we obtain the left-hand derivative

$$\begin{aligned}\text{L.H.D.} &= \lim_{h\to 0^+} \frac{f(-h) - f(0)}{-h} \\ &= \lim_{h\to 0^+} \frac{(-h)^2\sin(-1/h) - 0}{-h} \\ &= \lim_{h\to 0^+} \frac{-h^2\sin(1/h)}{-h} \quad (\because\ \sin(-t) = -\sin t) \\ &= \lim_{h\to 0^+} h\sin(1/h) \\ &= 0 \times \text{ a finite value keeps oscillating between } -1 \text{ and } 1 \\ &= 0.\end{aligned}$$

Putting $x = 0 + h$, we obtain the right-hand derivative

$$\begin{aligned}\text{R.H.D.} &= \lim_{h\to 0^+} \frac{f(h) - f(0)}{h} \\ &= \lim_{h\to 0^+} \frac{h^2 \sin(1/h) - 0}{h} \\ &= \lim_{h\to 0^+} h \sin(1/h) \\ &= 0 \times \text{ a finite value keeps oscillating between } -1 \text{ and } 1 \\ &= 0.\end{aligned}$$

Thus,

$$\text{L.H.D.} = \text{R.H.D.} = 0.$$

Therefore, f is differentiable at $x = 0$.
Since differentiability at a point implies continuity at that point, f is also continuous at $x = 0$.

$$\therefore \quad \boxed{f \text{ is continuous and differentiable at } 0.}$$

Indeterminate Forms and L'Hopital's Rule

Background: evaluating a limit

When evaluating a limit, first try direct substitution. For example,

$$\lim_{x\to 2} \frac{x+1}{x+3} = \frac{2+1}{2+3} = \frac{3}{5}.$$

However, direct substitution sometimes produces an expression that does not determine the limit. Such an expression is called an **indeterminate form**.

Indeterminate forms

The seven standard indeterminate forms are

$$\boxed{\frac{0}{0}, \qquad \frac{\infty}{\infty}, \qquad 0 \cdot \infty, \qquad \infty - \infty, \qquad 0^0, \qquad 1^\infty, \qquad \infty^0.}$$

These symbols describe the limiting behaviour of the component functions; they are not ordinary numerical calculations. For example, the form $0/0$ alone does not determine a limit:

$$\lim_{x\to 0} \frac{x}{x} = 1, \qquad \lim_{x\to 0} \frac{x^2}{x} = 0, \qquad \lim_{x\to 0} \frac{x}{x^2} \text{ does not exist as a two-sided limit.}$$

Important. A nonzero constant divided by a quantity tending to zero is not an indeterminate form. For example,

$$\lim_{x\to 0^+} \frac{1}{x} = +\infty.$$

L'Hopital's rule

Theorem 2.12 (L'Hopital's Rule)**.** *Suppose that f and g are differentiable in a deleted neighbourhood of a, with $g'(x) \neq 0$ there. Suppose also that either*

$$f(x) \to 0, \qquad g(x) \to 0,$$

or both $f(x)$ and $g(x)$ tend to signed infinities as $x \to a$.
If

$$\lim_{x \to a} \frac{f'(x)}{g'(x)} = L$$

exists, where L is finite or is $+\infty$ or $-\infty$, then

$$\boxed{\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)} = L.}$$

The corresponding rule also applies to one-sided limits and to limits as $x \to +\infty$ or $x \to -\infty$, under the analogous hypotheses.

How to apply the rule.

1. Check that the quotient has the form $\dfrac{0}{0}$ or $\dfrac{\infty}{\infty}$.
2. Differentiate the numerator and denominator separately. Do not use the quotient rule.
3. Evaluate the new limit.
4. If the new quotient is again indeterminate, the rule may be applied again, provided its hypotheses still hold.

Caution. If the derivative quotient has no limit, L'Hopital's rule gives no conclusion; the original limit may still exist.

Converting other indeterminate forms

1. **The form $0 \cdot \infty$.**

   Rewrite the product as a quotient:

   $$u(x)v(x) = \frac{u(x)}{1/v(x)} \qquad \text{or} \qquad u(x)v(x) = \frac{v(x)}{1/u(x)}.$$

   Choose the form that gives a simpler $\dfrac{0}{0}$ or $\dfrac{\infty}{\infty}$ limit.

2. **The form $\infty - \infty$.**

   Combine the terms using a common denominator, rationalise, or use an identity.

3. **The forms 0^0, 1^∞, and ∞^0.**

   Let

   $$y(x) = u(x)^{v(x)}, \qquad u(x) > 0.$$

Taking logarithms,

$$\log y(x) = \log\left(u(x)^{v(x)}\right) \implies \log y(x) = v(x)\log u(x).$$

If

$$\lim_{x\to a} v(x)\log u(x) = L,$$

then, by continuity of the exponential function,

$$\lim_{x\to a}\log y(x) = \lim_{x\to a} v(x)\log u(x) \implies \log\left(\lim_{x\to a} y(x)\right) = L \implies \lim_{x\to a} y(x) = e^L$$

Therefore,

$$\boxed{\lim_{x\to a} u(x)^{v(x)} = e^L.}$$

Thus, first evaluate the logarithmic limit and then exponentiate.

Useful standard limits and identities

$$\lim_{x\to 0}\frac{\sin x}{x} = 1 \text{ and } \lim_{x\to 0}\frac{\tan x}{x} = 1.$$

Problem 2.12. Evaluate: $\lim_{x\to 0}\dfrac{x-\sin x}{x^3}$.

(KUD: Feb-2025)(Question 8-QB)

Solution

Direct substitution gives the form $0/0$. Applying L'Hopital's rule repeatedly,

$$\begin{aligned}
\lim_{x\to 0}\frac{x-\sin x}{x^3} &= \lim_{x\to 0}\frac{1-\cos x}{3x^2} && (\because 0/0;\ \text{L'Hopital's rule})\\
&= \lim_{x\to 0}\frac{\sin x}{6x} && (\because 0/0;\ \text{L'Hopital's rule})\\
&= \lim_{x\to 0}\frac{\cos x}{6} && (\because 0/0;\ \text{L'Hopital's rule})\\
&= \boxed{\frac{1}{6}}.
\end{aligned}$$

Problem 2.13. Evaluate: $\lim_{x\to \pi/2}(\sec x - \tan x)$.

(Question 8-QB)

Solution

The two terms diverge, so their difference cannot be evaluated by subtracting their individual limits. Rewrite it as

$$\sec x - \tan x = \frac{1}{\cos x} - \frac{\sin x}{\cos x} = \frac{1-\sin x}{\cos x}.$$

This quotient has the form $0/0$. Hence,

$$\begin{aligned}\lim_{x\to\pi/2}(\sec x-\tan x) &= \lim_{x\to\pi/2}\frac{1-\sin x}{\cos x}\\ &= \lim_{x\to\pi/2}\frac{-\cos x}{-\sin x} \qquad (\because \text{L'Hopital's rule})\\ &= \boxed{0}.\end{aligned}$$

Problem 2.14. Evaluate: $\lim_{x\to 0^+} x\log x$.

(Question 8-QB)

Solution

The product has the form $0\cdot(-\infty)$.
Rewrite it as a quotient:

$$x\log x = \frac{\log x}{1/x}.$$

Therefore,

$$\begin{aligned}\lim_{x\to 0^+} x\log x &= \lim_{x\to 0^+}\frac{\log x}{1/x}\\ &= \lim_{x\to 0^+}\frac{1/x}{-1/x^2} \qquad (\because \text{L'Hopital's rule})\\ &= \lim_{x\to 0^+}(-x)\\ &= \boxed{0}.\end{aligned}$$

Problem 2.15. Evaluate: $\lim_{x\to 0}\dfrac{e^x-e^{-x}}{\sin x}$.

(Question 8-QB)

Solution

The quotient has the form $0/0$. Thus,

$$\begin{aligned}\lim_{x\to 0}\frac{e^x-e^{-x}}{\sin x} &= \lim_{x\to 0}\frac{e^x+e^{-x}}{\cos x} \qquad (\because \text{L'Hopital's rule})\\ &= \frac{1+1}{1}\\ &= \boxed{2}.\end{aligned}$$

Problem 2.16. Evaluate: $\lim_{x\to\infty}\dfrac{\log x}{x}$.

(Question 8-QB)

Solution

Both numerator and denominator tend to $+\infty$. Hence,

$$\begin{aligned}\lim_{x\to\infty}\frac{\log x}{x} &= \lim_{x\to\infty}\frac{1/x}{1} \qquad (\because \text{L'Hopital's rule})\\ &= \lim_{x\to\infty}\frac{1}{x}\\ &= \boxed{0}.\end{aligned}$$

Thus, $\log x$ grows more slowly than x.

Problem 2.17. Evaluate: $\lim_{x\to 0}(\operatorname{cosec} x - \cot x)$.

(Question 8-QB)

Solution

The two terms diverge, so their dierence cannot be evaluated by subtracting their individual limits. Rewrite it as:

$$\begin{aligned}\operatorname{cosec} x - \cot x &= \frac{1}{\sin x} - \frac{\cos x}{\sin x}\\ &= \frac{1-\cos x}{\sin x}.\end{aligned}$$

The resulting quotient has the form $0/0$. Therefore,

$$\begin{aligned}\lim_{x\to 0}(\operatorname{cosec} x - \cot x) &= \lim_{x\to 0}\frac{1-\cos x}{\sin x}\\ &= \lim_{x\to 0}\frac{\sin x}{\cos x} \qquad (\because \text{L'Hopital's rule})\\ &= \boxed{0}.\end{aligned}$$

Problem 2.18. Evaluate: $\lim_{x\to 0}\dfrac{1-\cos x}{x^2}$.

(Question 8-QB)

Solution

Direct substitution gives $0/0$. Hence,

$$\begin{aligned}\lim_{x\to 0}\frac{1-\cos x}{x^2} &= \lim_{x\to 0}\frac{\sin x}{2x} \qquad (\because 0/0;\ \text{L'Hopital's rule})\\ &= \lim_{x\to 0}\frac{\cos x}{2} \qquad (\because 0/0;\ \text{L'Hopital's rule})\\ &= \boxed{\frac{1}{2}}.\end{aligned}$$

Problem 2.19. Evaluate: $\lim_{x\to 0} \dfrac{e^x - e^{-x}}{\tan x}$.

(Question 8-QB)

Solution

The quotient has the form $0/0$. Therefore,

$$\begin{aligned}\lim_{x\to 0} \frac{e^x - e^{-x}}{\tan x} &= \lim_{x\to 0} \frac{e^x + e^{-x}}{\sec^2 x} && (\because 0/0;\ \text{L'Hopital's rule})\\ &= \frac{1+1}{1}\\ &= \boxed{2}.\end{aligned}$$

Problem 2.20. Evaluate: $\lim_{x\to 0} \dfrac{\tan x - \sin x}{x^3}$.

(Question 11-QB)

Solution

The quotient has the form $0/0$. Therefore,

$$\begin{aligned}\lim_{x\to 0} \frac{\tan x - \sin x}{x^3} &= \lim_{x\to 0} \frac{\sec^2 x - \cos x}{3x^2} && (\because 0/0;\ \text{L'Hopital's rule})\\ &= \lim_{x\to 0} \frac{2\sec^2 x \tan x + \sin x}{6x} && (\because 0/0;\ \text{L'Hopital's rule})\\ &= \lim_{x\to 0} \frac{4\sec^2 x \tan^2 x + 2\sec^4 x + \cos x}{6} && (\because 0/0;\ \text{L'Hopital's rule})\\ &= \frac{0+2+1}{6}\\ &= \boxed{\frac{1}{2}}.\end{aligned}$$

Problem 2.21. Evaluate: $\lim_{x\to 0^+} \dfrac{\log(\tan x)}{\log x}$.

(Question 11-QB)

Solution

Both logarithms tend to $-\infty$. Hence,

$$\begin{aligned}\lim_{x\to 0^+}\frac{\log(\tan x)}{\log x} &= \lim_{x\to 0^+}\frac{\sec^2 x/\tan x}{1/x} && (\because \text{L'Hopital's rule})\\ &= \lim_{x\to 0^+}\frac{x}{\sin x\cos x}\\ &= \lim_{x\to 0^+}\left(\frac{x}{\sin x}\right)\frac{1}{\cos x}\\ &= 1\cdot 1\\ &= \boxed{1}.\end{aligned}$$

Problem 2.22. Evaluate: $\lim_{x\to 0^+}\dfrac{\log(\sin x)}{\log(\sin 2x)}$.

(Question 11-QB)

Solution

Both numerator and denominator tend to $-\infty$. Therefore,

$$\begin{aligned}\lim_{x\to 0^+}\frac{\log(\sin x)}{\log(\sin 2x)} &= \lim_{x\to 0^+}\frac{\cot x}{2\cot 2x} && (\because \text{L'Hopital's rule})\\ &= \lim_{x\to 0^+}\frac{\cos x\sin 2x}{2\sin x\cos 2x}\\ &= \lim_{x\to 0^+}\frac{\cos^2 x}{\cos 2x} && (\because \sin 2x = 2\sin x\cos x)\\ &= \boxed{1}.\end{aligned}$$

Problem 2.23. Evaluate: $\lim_{x\to 0}\left(\dfrac{\tan x}{x}\right)^{1/x^2}$.

(Question 11-QB)

Solution

The expression has the form 1^∞. Let

$$y(x) = \left(\frac{\tan x}{x}\right)^{1/x^2}.$$

Taking logarithms,

$$\log y(x) = \frac{\log(\tan x/x)}{x^2}.$$

The quotient has the form $0/0$. Hence,

$$\begin{aligned}\lim_{x\to 0}\log y(x) &= \lim_{x\to 0}\frac{\sec^2 x/\tan x - 1/x}{2x} && (\because \text{L'Hopital's rule})\\ &= \lim_{x\to 0}\frac{1}{2x}\left(\frac{1}{\sin x\cos x}-\frac{1}{x}\right)\\ &= \lim_{x\to 0}\frac{1}{2x}\left(\frac{x-\sin x\cos x}{x\sin x\cos x}\right)\\ &= \lim_{x\to 0}\frac{x-\sin x\cos x}{2x^2\sin x\cos x}\\ &= \lim_{x\to 0}\left[\frac{x-\sin x\cos x}{x^3}\cdot\frac{x}{2\sin x\cos x}\right].\end{aligned}$$

Evaluate the first factor:

$$\begin{aligned}\lim_{x\to 0}\frac{x-\sin x\cos x}{x^3} &= \lim_{x\to 0}\frac{1-\cos 2x}{3x^2} && (\because \text{L'Hopital's rule})\\ &= \frac{2}{3}\lim_{x\to 0}\left(\frac{\sin x}{x}\right)^2\\ &= \frac{2}{3}.\end{aligned}$$

The second factor tends to $1/2$. Therefore,

$$\lim_{x\to 0}\log y(x) = \frac{2}{3}\cdot\frac{1}{2} = \frac{1}{3} \implies \log\left(\lim_{x\to 0} y(x)\right) = \frac{1}{3}.$$

Exponentiating,

$$\boxed{\lim_{x\to 0}\left(\frac{\tan x}{x}\right)^{1/x^2} = e^{1/3}.}$$

Problem 2.24. Evaluate: $\lim_{x\to 0}\left(\frac{\sin x}{x}\right)^{1/x^2}$.

(Question 11-QB)

Solution

The expression has the form 1^∞. Let

$$y(x) = \left(\frac{\sin x}{x}\right)^{1/x^2}.$$

Taking logarithms,

$$\log y(x) = \frac{\log(\sin x/x)}{x^2}.$$

Thus,

$$\begin{aligned}\lim_{x\to 0}\log y(x) &= \lim_{x\to 0}\frac{\frac{x}{\sin x}\cdot\frac{x\cos x-\sin x}{x^2}}{2x} && (\because 0/0;\ \text{L'Hopital's rule})\\ &= \lim_{x\to 0}\frac{x\cos x-\sin x}{2x^2\sin x}\\ &= \lim_{x\to 0}\left[\frac{x\cos x-\sin x}{x^3}\cdot\frac{x}{2\sin x}\right].\end{aligned}$$

Now,

$$\begin{aligned}\lim_{x\to 0}\frac{x\cos x-\sin x}{x^3} &= \lim_{x\to 0}\frac{-x\sin x}{3x^2} && (\because \text{L'Hopital's rule})\\ &= -\frac{1}{3}\lim_{x\to 0}\frac{\sin x}{x}\\ &= -\frac{1}{3}.\end{aligned}$$

Therefore,

$$\lim_{x\to 0}\log y(x) = -\frac{1}{3}\cdot\frac{1}{2} = -\frac{1}{6} \implies \log\left(\lim_{x\to 0} y(x)\right) = -\frac{1}{6}.$$

Exponentiating,

$$\boxed{\lim_{x\to 0}\left(\frac{\sin x}{x}\right)^{1/x^2} = e^{-1/6}.}$$

Problem 2.25. Evaluate: $\lim_{x\to\pi/2}(\sin x)^{\tan x}$.

(Question 11-QB)

Solution

The expression has the form 1^∞. Let

$$y(x) = (\sin x)^{\tan x}.$$

Taking logarithms,

$$\log y(x) = \tan x\log(\sin x) = \frac{\log(\sin x)}{\cot x}.$$

The quotient has the form $0/0$. Hence,

$$\begin{aligned}\lim_{x\to\pi/2}\log y(x) &= \lim_{x\to\pi/2}\frac{\cot x}{-\operatorname{cosec}^2 x} && (\because \text{L'Hopital's rule})\\ &= \lim_{x\to\pi/2}(-\sin x\cos x)\\ &= 0.\end{aligned}$$

Therefore,

$$\boxed{\lim_{x\to\pi/2}(\sin x)^{\tan x} = e^0 = 1.}$$

Problem 2.26. Evaluate: $\lim_{x\to 0^+} (\sin x)^{\tan x}$.

(Question 11-QB)

Solution

The expression has the form 0^0. Let

$$y(x) = (\sin x)^{\tan x}, \qquad x > 0.$$

Then

$$\log y(x) = \frac{\log(\sin x)}{\cot x}.$$

The numerator tends to $-\infty$, while the denominator tends to $+\infty$. Therefore,

$$\begin{aligned}\lim_{x\to 0^+} \log y(x) &= \lim_{x\to 0^+} \frac{\cot x}{-\operatorname{cosec}^2 x} \qquad (\because \text{L'Hopital's rule})\\ &= \lim_{x\to 0^+} (-\sin x \cos x)\\ &= 0.\end{aligned}$$

Hence,

$$\boxed{\lim_{x\to 0^+} (\sin x)^{\tan x} = e^0 = 1.}$$

Problem 2.27. Evaluate: $\lim_{x\to 0} \left(\frac{1}{x^2} - \cot^2 x\right)$.

(Question 11-QB)

Solution

The expression has the form $\infty - \infty$. Combining the terms and factoring,

$$\begin{aligned}\frac{1}{x^2} - \cot^2 x &= \frac{\sin^2 x - x^2\cos^2 x}{x^2 \sin^2 x}\\ &= \frac{(\sin x - x\cos x)(\sin x + x\cos x)}{x^2 \sin^2 x}\\ &= \frac{\sin x - x\cos x}{x^3} \cdot \frac{\sin x + x\cos x}{x} \cdot \frac{x^2}{\sin^2 x}.\end{aligned}$$

For the first factor,

$$\begin{aligned}\lim_{x\to 0} \frac{\sin x - x\cos x}{x^3} &= \lim_{x\to 0} \frac{x \sin x}{3x^2} \qquad (\because 0/0;\ \text{L'Hopital's rule})\\ &= \frac{1}{3}.\end{aligned}$$

For the remaining factors,

$$\lim_{x\to 0} \frac{\sin x + x\cos x}{x} = \lim_{x\to 0} \left(\frac{\sin x}{x} + \cos x\right) = 2,$$

and

$$\lim_{x\to 0}\frac{x^2}{\sin^2 x}=1.$$

Therefore,

$$\boxed{\lim_{x\to 0}\left(\frac{1}{x^2}-\cot^2 x\right)=\frac{1}{3}\cdot 2\cdot 1=\frac{2}{3}.}$$

Problem 2.28. Evaluate: $\lim_{x\to 0^+}(\cot x)^{\sin x}$.

(Question 11-QB)

Solution

The expression has the form ∞^0. Let

$$y(x)=(\cot x)^{\sin x}.$$

Taking logarithms,

$$\log y(x)=\sin x\log(\cot x)=\frac{\log(\cot x)}{\operatorname{cosec} x}.$$

Both numerator and denominator tend to $+\infty$. Hence,

$$\begin{aligned}\lim_{x\to 0^+}\log y(x)&=\lim_{x\to 0^+}\frac{-\operatorname{cosec}^2 x/\cot x}{-\operatorname{cosec} x\cot x}\qquad(\because \text{L'Hopital's rule})\\&=\lim_{x\to 0^+}\frac{\operatorname{cosec} x}{\cot^2 x}\\&=\lim_{x\to 0^+}\frac{\sin x}{\cos^2 x}\\&=0.\end{aligned}$$

Therefore,

$$\boxed{\lim_{x\to 0^+}(\cot x)^{\sin x}=e^0=1.}$$

Problem 2.29. Evaluate: $\lim_{x\to 0^+}\sin x\log x$.

(Question 11-QB)

Solution

The product has the form $0\cdot(-\infty)$. Write

$$\sin x\log x=\frac{\log x}{\operatorname{cosec}\ x}.$$

Then

$$\begin{aligned}\lim_{x\to 0^+} \sin x \log x &= \lim_{x\to 0^+} \frac{1/x}{-\operatorname{cosec} x \cot x} && (\because \text{L'Hopital's rule})\\ &= -\lim_{x\to 0^+} \frac{\sin^2 x}{x\cos x}\\ &= -\lim_{x\to 0^+} \left(\frac{\sin x}{x}\right)\tan x\\ &= -1\cdot 0\\ &= \boxed{0}.\end{aligned}$$

Problem 2.30. Evaluate: $\lim_{x\to 0}\left(\frac{1}{x} - \cot x\right)$.

(Question 11-QB)

Solution

Combining the terms,

$$\frac{1}{x} - \cot x = \frac{\sin x - x\cos x}{x\sin x}.$$

The quotient has the form $0/0$. Therefore,

$$\begin{aligned}\lim_{x\to 0}\left(\frac{1}{x} - \cot x\right) &= \lim_{x\to 0} \frac{x\sin x}{\sin x + x\cos x} && (\because \text{L'Hopital's rule})\\ &= \lim_{x\to 0} \frac{\sin x}{\sin x/x + \cos x} && (\because \text{divide by } x \neq 0)\\ &= \frac{0}{1+1}\\ &= \boxed{0}.\end{aligned}$$

Problem 2.31. Evaluate: $\lim_{x\to 0}\left(\frac{1}{x} - \frac{1}{e^x - 1}\right)$.

(Question 11-QB)

Solution

Combining the terms,

$$\frac{1}{x} - \frac{1}{e^x - 1} = \frac{e^x - 1 - x}{x(e^x - 1)}.$$

The quotient has the form $0/0$. Hence,

$$\begin{aligned}\lim_{x\to 0}\left(\frac{1}{x}-\frac{1}{e^x-1}\right) &= \lim_{x\to 0}\frac{e^x-1}{e^x-1+xe^x} && (\because \text{L'Hopital's rule})\\ &= \lim_{x\to 0}\frac{e^x}{2e^x+xe^x} && (\because 0/0;\ \text{L'Hopital's rule})\\ &= \frac{1}{2}\\ &= \boxed{\frac{1}{2}}.\end{aligned}$$

Problem 2.32. Evaluate: $\lim_{x\to 1} x^{1/(x-1)}$.

(Question 11-QB)

Solution

The expression has the form 1^∞. Let

$$y(x) = x^{1/(x-1)}.$$

Taking logarithms,

$$\log y(x) = \frac{\log x}{x-1}.$$

This quotient has the form $0/0$. Therefore,

$$\begin{aligned}\lim_{x\to 1}\log y(x) &= \lim_{x\to 1}\frac{1/x}{1} && (\because \text{L'Hopital's rule})\\ &= 1.\end{aligned}$$

Exponentiating,

$$\boxed{\lim_{x\to 1} x^{1/(x-1)} = e.}$$

Problem 2.33. Evaluate: $\lim_{x\to 0}\dfrac{x-\tan x}{x^3}$.

(KUD: Jan-2026)(Question 11-QB)

Solution

The quotient has the form $0/0$. Thus,

$$\begin{aligned}\lim_{x\to 0}\frac{x-\tan x}{x^3} &= \lim_{x\to 0}\frac{1-\sec^2 x}{3x^2} && (\because \text{L'Hopital's rule})\\ &= -\frac{1}{3}\lim_{x\to 0}\frac{\tan^2 x}{x^2} && (\because \sec^2 x - 1 = \tan^2 x)\\ &= -\frac{1}{3}\left(\lim_{x\to 0}\frac{\tan x}{x}\right)^2\\ &= \boxed{-\frac{1}{3}}.\end{aligned}$$

Problem 2.34. Evaluate: $\lim_{x\to 0}\dfrac{e^x - e^{-x} - 2x}{x^2\sin x}$.

(Question 11-QB)

Solution

The quotient has the form $0/0$. Therefore,

$$\begin{aligned}&\lim_{x\to 0}\frac{e^x - e^{-x} - 2x}{x^2\sin x}\\ &= \lim_{x\to 0}\frac{e^x + e^{-x} - 2}{2x\sin x + x^2\cos x} && (\because 0/0;\ \text{L'Hopital's rule})\\ &= \lim_{x\to 0}\frac{e^x - e^{-x}}{2\sin x + 4x\cos x - x^2\sin x} && (\because 0/0;\ \text{L'Hopital's rule})\\ &= \lim_{x\to 0}\frac{e^x + e^{-x}}{6\cos x - 6x\sin x - x^2\cos x} && (\because 0/0;\ \text{L'Hopital's rule})\\ &= \frac{1+1}{6}\\ &= \boxed{\frac{1}{3}}.\end{aligned}$$

Problem 2.35. Evaluate: $\lim_{x\to 0}\left(\dfrac{a^x + b^x}{2}\right)^{1/x}$, $\quad a, b > 0$.

(KUD: Jan-2026)

Solution

The expression has the form 1^∞. Let

$$y = \left(\frac{a^x + b^x}{2}\right)^{1/x}.$$

Taking logarithms,

$$\log y = \frac{\log\left(\frac{a^x + b^x}{2}\right)}{x}.$$

This quotient has the form $0/0$. Hence,

$$\begin{aligned}\lim_{x\to 0} \log y &= \lim_{x\to 0} \frac{\log\left(\frac{a^x + b^x}{2}\right)}{x} \\ &= \lim_{x\to 0} \frac{a^x \log a + b^x \log b}{a^x + b^x} \qquad (\because \text{L'Hopital's rule}) \\ &= \frac{\log a + \log b}{2} \\ &= \log \sqrt{ab}.\end{aligned}$$

Therefore,

$$\begin{aligned}\lim_{x\to 0} \left(\frac{a^x + b^x}{2}\right)^{1/x} &= e^{\log \sqrt{ab}} \qquad (\because \text{continuity of the exponential function}) \\ &= \boxed{\sqrt{ab}}.\end{aligned}$$

Problem 2.36. Evaluate: $\lim_{x\to 0}(\cos x)^{1/x}$.

(KUD: Feb-2025)

Solution

The expression has the form 1^∞. Let

$$y = (\cos x)^{1/x}.$$

Taking logarithms,

$$\log y = \frac{\log(\cos x)}{x}.$$

This quotient has the form $0/0$. Hence,

$$\begin{aligned}\lim_{x\to 0} \log y &= \lim_{x\to 0} \frac{\log(\cos x)}{x} \\ &= \lim_{x\to 0} \frac{-\tan x}{1} \qquad (\because \text{L'Hopital's rule}) \\ &= 0.\end{aligned}$$

Therefore,

$$\begin{aligned}\lim_{x\to 0}(\cos x)^{1/x} &= e^0 \qquad (\because \text{continuity of the exponential function}) \\ &= \boxed{1}.\end{aligned}$$

3 Unit-IV: Successive Differentiation and MVTs

Mean Value Theorems

Theorem 3.1 (Rolle's Theorem). *Let $f : [a, b] \to \mathbb{R}$ satisfy the following conditions:*
(i). *f is continuous on $[a, b]$*
(ii). *f is differentiable on (a, b)*
(iii). *$f(a) = f(b)$.*
Then there exists at least one point $c \in (a, b)$ such that $f'(c) = 0$.
(KUD: Feb-2025)(Question 13-QB)

Proof

Since f is continuous on $[a, b]$, it attains its supremum and infimum values. Let

$$M = \sup_{x\in[a,b]} f(x) = f(\alpha) \text{ and } m = \inf_{x\in[a,b]} f(x) = f(\beta) \text{ for some } \alpha, \beta \in [a, b].$$

We consider two cases.
Case 1: $M = m$**.** For every $x \in [a, b]$, $m \le f(x) \le M$. Thus $f(x) = M$ throughout $[a, b]$, so f is constant. Consequently, $f'(x) = 0$ for every $x \in (a, b)$.
Case 2: $M \neq m$**.** Put $k = f(a) = f(b)$. At least one of M and m differs from k. Suppose first that $M \neq k$. Since $M = f(\alpha) \neq f(a) = k$ and $M = f(\alpha) \neq f(b) = k \implies \alpha \neq a$ and $\alpha \neq b$, so we have $\alpha \in (a, b)$.
For sufficiently small $h > 0$, both $c - h$ and $c + h$ belong to $[a, b]$. Since $f(\alpha)$ is supremum,

$$f(\alpha - h) - f(\alpha) \le 0 \text{ and } f(\alpha + h) - f(\alpha) \le 0.$$

Dividing by $-h < 0$ and $h > 0$, respectively, gives

$$\frac{f(\alpha - h) - f(\alpha)}{-h} \ge 0 \text{ and } \frac{f(\alpha + h) - f(\alpha)}{h} \le 0.$$

Taking limits as $h \to 0^+$,

$$Lf'(\alpha) = \lim_{h\to 0^+} \frac{f(\alpha - h) - f(\alpha)}{-h} \ge 0 \text{ and } Rf'(\alpha) = \lim_{h\to 0^+} \frac{f(\alpha + h) - f(\alpha)}{h} \le 0.$$

Because f is differentiable at α,

$$Lf'(\alpha) = Rf'(\alpha) = f'(\alpha) \implies f'(\alpha) = 0.$$

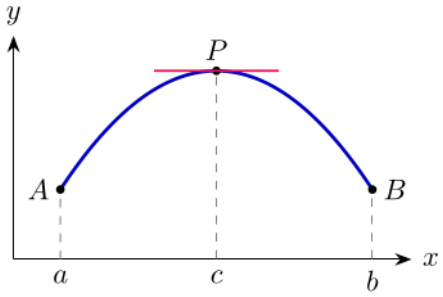
Similarly, if $m \neq k$, one can prove that $f'(\beta) = 0$. Thus, there exists at least one point $c \in (a, b)$ such that $f'(c) = 0$. □

Geometrical interpretation

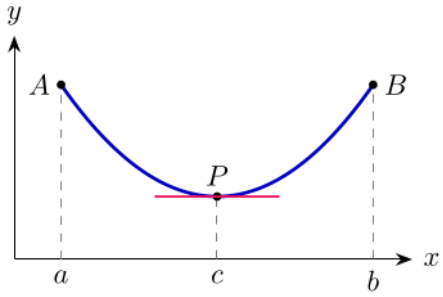
Let $A = (a, f(a))$ and $B = (b, f(b))$ be points on the graph $y = f(x)$. Since $f(a) = f(b)$, the points A and B have the same height, and the chord AB is horizontal.

Rolle's theorem guarantees at least one point $P = (c, f(c))$, with $a < c < b$, at which the tangent is parallel to the x-axis. Its slope is $f'(c) = 0$.

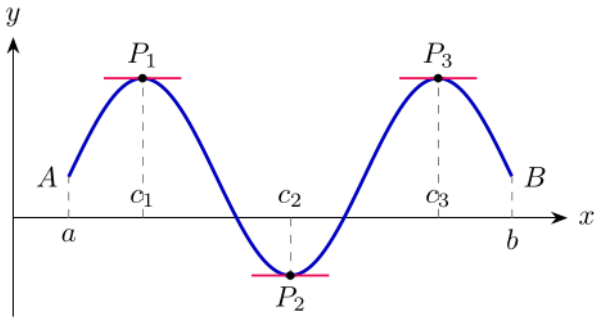
The following diagrams illustrate a maximum, a minimum, and several possible points with horizontal tangents.



An interior maximum.



An interior minimum.



There may be more than one point satisfying $f'(c) = 0$.

Example 3.1. Verify Rolle's theorem for $f(x) = x^2 - 6x + 8$ on $[2, 4]$.

Solution

As a polynomial, f is continuous on $[2, 4]$ and differentiable on $(2, 4)$. Also,

$$f(2) = 4 - 12 + 8 = 0 \text{ and } f(4) = 16 - 24 + 8 = 0.$$

Thus all three hypotheses are satisfied. Since $f'(x) = 2x - 6$,

$$f'(c) = 0 \implies 2c - 6 = 0 \implies 2c = 6 \implies \boxed{c = 3 \in (2, 4).}$$

Hence Rolle's theorem is verified.

The Lagrange's mean value theorem is also called the *first mean value theorem of differential calculus*.

Theorem 3.2 (Lagrange's Mean Value Theorem). *Let $f : [a, b] \to \mathbb{R}$ satisfy the following conditions:*

(i). *f is continuous on $[a, b]$*

(ii). *f is differentiable on (a, b)*

Then there exists at least one $c \in (a, b)$ such that $f'(c) = \dfrac{f(b) - f(a)}{b - a}$.

(KUD: Jan-2026, Feb-2025)(Question 14-QB)

Proof

We construct an auxiliary function to which Rolle's theorem applies.
Let $\phi(x) = f(x) + Ax$, where A is chosen so that $\phi(a) = \phi(b)$. Thus,

$$\begin{aligned} f(a) + Aa &= f(b) + Ab, \\ A(a - b) &= f(b) - f(a), \\ A &= -\frac{f(b) - f(a)}{b - a}. \end{aligned}$$

Now ϕ is continuous on $[a, b]$ and differentiable on (a, b), being the sum of f and a linear function. By the choice of A, $\phi(a) = \phi(b)$.
Rolle's theorem therefore gives $c \in (a, b)$ with $\phi'(c) = 0$. Since $\phi'(x) = f'(x) + A$,

$$\phi'(c) = 0 \implies f'(c) + A = 0 \implies f'(c) = -A = \frac{f(b) - f(a)}{b - a}.$$

This proves the theorem. □

Relation to Rolle's theorem

Note. If $f(a) = f(b)$, Lagrange's conclusion becomes $f'(c) = 0$. Thus Rolle's theorem is the equal-endpoint special case of Lagrange's mean value theorem.

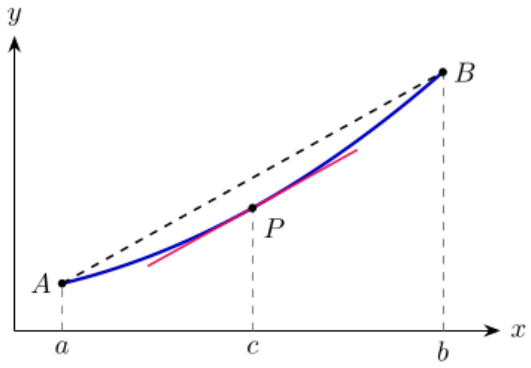
Geometrical interpretation

Let $A = (a, f(a))$ and $B = (b, f(b))$. The slope of the chord joining A and B is

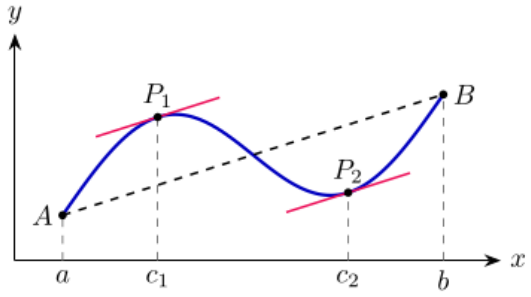
$$\frac{f(b) - f(a)}{b - a}.$$

At an interior point $P = (c, f(c))$, the tangent has slope $f'(c)$. The theorem guarantees at least one such point where the tangent is parallel to the chord AB.

In the diagrams below, the dashed line is the chord and the orange line segments are tangents parallel to it.



A tangent parallel to the chord AB.



More than one point may satisfy the conclusion.

Example 3.2. Verify Lagrange's mean value theorem for $f(x) = (x+1)(x+2)$ on $[1,2]$.

(Question 16-QB)

Solution

The function $f(x) = (x+1)(x+2) = x^2 + 3x + 2$ is a polynomial. Therefore,
(i). f is continuous on $[1,2]$
(ii). f is differentiable on $(1,2)$.
Thus, both hypotheses of Lagrange's mean value theorem are satisfied. Hence, there exists $c \in (1,2)$ such that

$$f'(c) = \frac{f(2) - f(1)}{2 - 1}.$$

Now,

$$f(1) = (1+1)(1+2) = 6 \text{ and } f(2) = (2+1)(2+2) = 12,$$

and

$$f'(x) = 2x + 3.$$

Therefore,

$$f'(c) = \frac{f(2) - f(1)}{2 - 1} \implies 2c + 3 = \frac{12 - 6}{2 - 1} = 6 \implies 2c = 3 \implies \boxed{c = \frac{3}{2}.}$$

Since $\frac{3}{2} \in (1,2)$, Lagrange's mean value theorem is verified.

The Cauchy's mean value theorem is also called the *second mean value theorem of differential calculus.*

Theorem 3.3 (Cauchy's Mean Value Theorem). *Let $f, g : [a, b] \to \mathbb{R}$ satisfy the following conditions:*

(i). *f and g are continuous on $[a, b]$*
(ii). *f and g are differentiable on (a, b)*
(iii). *$g'(x) \neq 0$ for every $x \in (a, b)$.*

Then there exists at least one $c \in (a, b)$ such that $\dfrac{f(b) - f(a)}{g(b) - g(a)} = \dfrac{f'(c)}{g'(c)}$.

(KUD: Jan-2026)(Question 15-QB)

Proof

We shall first show that $g(b) - g(a) \neq 0$. Suppose $g(a) = g(b)$. Then g satisfies all the hypotheses of Rolle's theorem, so there exists $d \in (a, b)$ with $g'(d) = 0$. This contradicts hypothesis (iii). Therefore, $g(a) \neq g(b)$.
Now, construct an auxiliary function.
Let $\phi(x) = f(x) + Ag(x)$, where A is chosen so that $\phi(a) = \phi(b)$. Thus,

$$\begin{aligned} f(a) + Ag(a) &= f(b) + Ag(b), \\ A[g(a) - g(b)] &= f(b) - f(a), \\ A &= -\frac{f(b) - f(a)}{g(b) - g(a)}. \end{aligned}$$

The function ϕ is continuous on $[a, b]$, differentiable on (a, b), being the sum of f and Ag, and satisfies $\phi(a) = \phi(b)$. Rolle's theorem therefore gives a point $c \in (a, b)$ such that

$$\phi'(c) = 0 \implies f'(c) + Ag'(c) = 0 \implies f'(c) = -Ag'(c).$$

Since $g'(c) \neq 0$,

$$\frac{f'(c)}{g'(c)} = -A = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

This proves the theorem. □

Relation to Lagrange's theorem

Note. Set $g(x) = x$. Then $g'(x) = 1 \neq 0$, and $g(b) - g(a) = b - a$.
Cauchy's theorem becomes $\dfrac{f(b) - f(a)}{b - a} = f'(c)$, which is Lagrange's mean value theorem.

Example 3.3. Verify Cauchy's mean value theorem for $f(x) = \sin x$ and $g(x) = \cos x$ on $\left[0, \frac{\pi}{2}\right]$.

(Question 17-QB)

Solution

The functions $f(x) = \sin x$ and $g(x) = \cos x$ satisfy:

(i). Both are continuous on $\left[0, \frac{\pi}{2}\right]$.

(ii). Both are differentiable on $\left(0, \frac{\pi}{2}\right)$.

(iii). $g'(x) = -\sin x \neq 0$ for every $x \in \left(0, \frac{\pi}{2}\right)$.

Thus, all hypotheses of Cauchy's mean value theorem are satisfied. Hence, there exists $c \in \left(0, \frac{\pi}{2}\right)$ such that

$$\frac{f\left(\frac{\pi}{2}\right) - f(0)}{g\left(\frac{\pi}{2}\right) - g(0)} = \frac{f'(c)}{g'(c)}.$$

Now,

$$f(0) = \sin 0 = 0,\ f\left(\frac{\pi}{2}\right) = \sin\frac{\pi}{2} = 1,\ g(0) = \cos 0 = 1,\ g\left(\frac{\pi}{2}\right) = \cos\frac{\pi}{2} = 0,$$

and

$$f'(x) = \cos x \text{ and } g'(x) = -\sin x.$$

Therefore,

$$\frac{f\left(\frac{\pi}{2}\right) - f(0)}{g\left(\frac{\pi}{2}\right) - g(0)} = \frac{f'(c)}{g'(c)} \implies \frac{1-0}{0-1} = \frac{\cos c}{-\sin c} \implies \cos c = \sin c \implies \tan c = 1.$$

Since $c \in \left(0, \frac{\pi}{2}\right)$, we obtain

$$\boxed{c = \frac{\pi}{4}.}$$

Thus, Cauchy's mean value theorem is verified.

Taylor's Theorem with Schlömilch–Roche Remainder

Theorem 3.4 (Taylor's Theorem). *Let $h > 0$, and let n be a positive integer. Suppose that $f, f', \ldots, f^{(n-1)}$ are continuous on $[a, a+h]$ and that $f^{(n)}$ exists on $(a, a+h)$. Then, for any positive integer p, there exists $\theta \in (0,1)$ such that*

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \cdots + \frac{h^{n-1}}{(n-1)!}f^{(n-1)}(a) + R_n,$$

where

$$R_n = \frac{h^n(1-\theta)^{n-p}}{p(n-1)!}f^{(n)}(a+\theta h).$$

*This is the **Schlömilch–Roche form of the remainder** after n terms.*

(Question 7-QB)

Taylor's and Maclaurin's Infinite Series

Suppose that f has continuous derivatives of every order on an open interval containing a and $a+h$. For every positive integer n, Taylor's theorem gives

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \cdots + \frac{h^{n-1}}{(n-1)!}f^{(n-1)}(a) + R_n(h),$$

where,

$$R_n(h) = \frac{h^n(1-\theta)^{n-p}}{p(n-1)!}f^{(n)}(a+\theta h).$$

The number θ may depend on n and h.

Let

$$S_n(h) = f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \cdots + \frac{h^{n-1}}{(n-1)!}f^{(n-1)}(a).$$

Thus,

$$f(a+h) = S_n(h) + R_n(h).$$

For a fixed h, if $\lim_{n\to\infty} R_n(h) = 0$, then $\lim_{n\to\infty} S_n(h) = f(a+h)$. Consequently,

$$\boxed{f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \cdots + \frac{h^{n-1}}{(n-1)!}f^{(n-1)}(a) + \cdots .}$$

Thus, taking $x = a+h$, we obtain

Definition 3.1 (Taylor Series)**.** The series

$$\boxed{f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \frac{(x-a)^3}{3!}f'''(a) + \cdots .}$$

is called the **Taylor series of f about a**, where $f^{(0)}(a) = f(a)$.
At any point x where the Taylor remainder tends to zero,

Definition 3.2 (Maclaurin Series)**.** The Taylor series about $a = 0$ is called the **Maclaurin series of f**:

$$\boxed{f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + +\frac{x^3}{3!}f'''(0) + + \cdots .}$$

Whenever the corresponding remainder tends to zero,

Remark. The existence of continuous derivatives of every order does not, by itself, guarantee that the Taylor series represents the function. We must also establish that

$$R_n \longrightarrow 0 \qquad \text{as } n \to \infty.$$

Any valid form of Taylor's remainder may be used to verify this condition.

Problem 3.1. Expand $f(x) = e^x$ in powers of x using Maclaurin's expansion.
(KUD: Feb-2025)(Question 8-QB)

Solution

The given function is $f(x) = e^x$.

Maclaurin's expansion is $f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \frac{x^4}{4!}f^{iv}(0) + \cdots$.

The derivatives and their values at $x = 0$ are

Function	Expression	Value at $x = 0$
f	e^x	1
f'	e^x	1
f''	e^x	1
f'''	e^x	1
f^{iv}	e^x	1

Substitute in Maclaurin's expansion:

$$\begin{aligned} e^x &= 1 + x(1) + \frac{x^2}{2!}(1) + \frac{x^3}{3!}(1) + \frac{x^4}{4!}(1) + \cdots \\ &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots . \end{aligned}$$

Therefore,

$$\boxed{e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots , \qquad x \in \mathbb{R}.}$$

Problem 3.2. Expand $f(x) = \cos x$ in powers of x using Maclaurin's expansion.
(KUD: Jan-2026)(Question 8-QB)

Solution

The given function is $f(x) = \cos x$.

Maclaurin's expansion is $f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \frac{x^4}{4!}f^{iv}(0) + \cdots$.

The derivatives and their values at $x = 0$ are

Function	Expression	Value at $x = 0$
f	$\cos x$	1
f'	$-\sin x$	0
f''	$-\cos x$	-1
f'''	$\sin x$	0
f^{iv}	$\cos x$	1

Substitute in Maclaurin's expansion:

$$\begin{aligned}\cos x &= 1 + x(0) + \frac{x^2}{2!}(-1) + \frac{x^3}{3!}(0) + \frac{x^4}{4!}(1) + \cdots \\ &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots .\end{aligned}$$

Therefore,

$$\boxed{\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots , \qquad x \in \mathbb{R}.}$$

Problem 3.3. Expand $f(x) = \sin x$ in powers of x using Maclaurin's expansion.

(Question 8-QB)

Solution

The given function is $f(x) = \sin x$.

Maclaurin's expansion is $f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \frac{x^4}{4!}f^{iv}(0) + \cdots$.

The derivatives and their values at $x = 0$ are

Function	Expression	Value at $x = 0$
f	$\sin x$	0
f'	$\cos x$	1
f''	$-\sin x$	0
f'''	$-\cos x$	-1
f^{iv}	$\sin x$	0

Substitute in Maclaurin's expansion:

$$\begin{aligned}\sin x &= 0 + x(1) + \frac{x^2}{2!}(0) + \frac{x^3}{3!}(-1) + \frac{x^4}{4!}(0) + \cdots \\ &= x - \frac{x^3}{3!} + \cdots .\end{aligned}$$

Therefore,

$$\boxed{\sin x = x - \frac{x^3}{3!} + \cdots , \qquad x \in \mathbb{R}.}$$

Problem 3.4. Using Maclaurin's theorem, prove that

$$\log(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots , \qquad -1 < x \leq 1.$$

(Question 18-QB)

Solution

The given function is $f(x) = \log(1+x)$.

Maclaurin's expansion is $f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \cdots$.

The derivatives and their values at $x = 0$ are

Function	Expression	Value at $x = 0$
f	$\log(1+x)$	0
f'	$\frac{1}{1+x}$	1
f''	$-\frac{1}{(1+x)^2}$	-1
f'''	$\frac{2!}{(1+x)^3}$	$2!$

Substitute in Maclaurin's expansion:

$$\begin{aligned}\log(1+x) &= 0 + x(1) + \frac{x^2}{2!}(-1) + \frac{x^3}{3!}(2!) + \cdots \\ &= x - \frac{x^2}{2} + \frac{x^3}{3} + \cdots,\end{aligned}$$

Therefore,

$$\boxed{\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots, \qquad -1 < x \leq 1.}$$

Problem 3.5. Using Maclaurin's series, prove that

$$\log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \cdots.$$

(Question 18-QB)

Solution

The given function is $f(x) = \log(1-x)$.

Maclaurin's expansion is $f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \cdots$.

The derivatives and their values at $x = 0$ are

Function	Expression	Value at $x = 0$
f	$\log(1-x)$	0
f'	$-\frac{1}{1-x}$	-1
f''	$-\frac{1}{(1-x)^2}$	-1
f'''	$-\frac{2!}{(1-x)^3}$	$-2!$

Substitute in Maclaurin's expansion:

$$\begin{aligned}\log(1-x) &= 0 + x(-1) + \frac{x^2}{2!}(-1) + \frac{x^3}{3!}(-2!) + \cdots \\ &= -x - \frac{x^2}{2} - \frac{x^3}{3} - \cdots .\end{aligned}$$

Therefore

$$\boxed{\log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \cdots , \qquad -1 \le x < 1.}$$

Problem 3.6. Using Maclaurin's theorem, obtain

$$\log(1+\sin x) = x - \frac{x^2}{2} + \frac{x^3}{6} - \frac{x^4}{12} + \cdots .$$

(Question 19-QB)

Solution

The given function is $f(x) = \log(1+\sin x)$.

Maclaurin's expansion is $f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \frac{x^4}{4!}f^{(4)}(0) + \cdots$.

Differentiating directly,

$$f'(x) = \frac{\cos x}{1+\sin x}.$$

Using the quotient rule,

$$\begin{aligned}f''(x) &= \frac{-\sin x(1+\sin x) - \cos^2 x}{(1+\sin x)^2} \\ &= \frac{-(1+\sin x)}{(1+\sin x)^2} \\ &= -\frac{1}{1+\sin x}.\end{aligned}$$

Differentiating again,

$$f'''(x) = \frac{\cos x}{(1+\sin x)^2},$$

and

$$\begin{aligned} f^{iv}(x) &= -\frac{\sin x}{(1+\sin x)^2} - \frac{2\cos^2 x}{(1+\sin x)^3} \\ &= \frac{-\sin x(1+\sin x) - 2\cos^2 x}{(1+\sin x)^3} \\ &= \frac{\sin^2 x - \sin x - 2}{(1+\sin x)^3} \\ &= \frac{(\sin x - 2)(1+\sin x)}{(1+\sin x)^3} \\ &= \frac{\sin x - 2}{(1+\sin x)^2}. \end{aligned}$$

Thus,

Function	Expression	Value at $x = 0$
f	$\log(1+\sin x)$	0
f'	$\dfrac{\cos x}{1+\sin x}$	1
f''	$-\dfrac{1}{1+\sin x}$	-1
f'''	$\dfrac{\cos x}{(1+\sin x)^2}$	1
f^{iv}	$\dfrac{\sin x - 2}{(1+\sin x)^2}$	-2

Substitute in Maclaurin's expansion:

$$\begin{aligned} \log(1+\sin x) &= 0 + x(1) + \frac{x^2}{2!}(-1) + \frac{x^3}{3!}(1) + \frac{x^4}{4!}(-2) + \cdots \\ &= x - \frac{x^2}{2} + \frac{x^3}{6} - \frac{x^4}{12} + \cdots . \end{aligned}$$

Therefore,

$$\boxed{\log(1+\sin x) = x - \frac{x^2}{2} + \frac{x^3}{6} - \frac{x^4}{12} + \cdots .}$$

Problem 3.7. Expand $\log(1+e^x)$ up to the term containing x^4, using Maclaurin's theorem.

(Question 20-QB)

Solution

The given function is $f(x) = \log(1+e^x)$.

Maclaurin's expansion is $f(x) = f(0) + xf'(0) + \dfrac{x^2}{2!}f''(0) + \dfrac{x^3}{3!}f'''(0) + \dfrac{x^4}{4!}f^{(4)}(0) + \cdots$.

Differentiating directly,

$$f'(x) = \frac{e^x}{1+e^x},$$

and

$$f''(x) = \frac{e^x(1+e^x) - e^{2x}}{(1+e^x)^2} = \frac{e^x}{(1+e^x)^2}.$$

Next,

$$f'''(x) = \frac{e^x}{(1+e^x)^2} - \frac{2e^{2x}}{(1+e^x)^3} = \frac{e^x(1-e^x)}{(1+e^x)^3}.$$

Finally,

$$f^{iv}(x) = \frac{e^x - 2e^{2x}}{(1+e^x)^3} - \frac{3e^x(e^x - e^{2x})}{(1+e^x)^4} = \frac{(e^x - 2e^{2x})(1+e^x) - 3e^x(e^x - e^{2x})}{(1+e^x)^4} = \frac{e^x - 4e^{2x} + e^{3x}}{(1+e^x)^4}.$$

Thus,

Function	Expression	Value at $x = 0$
f	$\log(1+e^x)$	$\log 2$
f'	$\frac{e^x}{1+e^x}$	$\frac{1}{2}$
f''	$\frac{e^x}{(1+e^x)^2}$	$\frac{1}{4}$
f'''	$\frac{e^x(1-e^x)}{(1+e^x)^3}$	0
f^{iv}	$\frac{e^x - 4e^{2x} + e^{3x}}{(1+e^x)^4}$	$-\frac{1}{8}$

Substitute in Maclaurin's expansion:

$$\log(1+e^x) = \log 2 + x\left(\frac{1}{2}\right) + \frac{x^2}{2!}\left(\frac{1}{4}\right) + \frac{x^3}{3!}(0) + \frac{x^4}{4!}\left(-\frac{1}{8}\right) + \cdots = \log 2 + \frac{x}{2} + \frac{x^2}{8} - \frac{x^4}{192} + \cdots.$$

Therefore,

$$\boxed{\log(1+e^x) = \log 2 + \frac{x}{2} + \frac{x^2}{8} - \frac{x^4}{192} + \cdots.}$$

Problem 3.8. Expand $e^{\sin x}$ up to the term containing x^4, using Maclaurin's expansion.

(Question 21-QB)

Solution

The given function is $f(x) = e^{\sin x}$.

Maclaurin's expansion is $f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \frac{x^4}{4!}f^{(4)}(0) + \cdots$.

Differentiating directly,

$$f'(x) = e^{\sin x}\cos x.$$

Using the product rule,

$$\begin{aligned} f''(x) &= e^{\sin x}\cos^2 x - e^{\sin x}\sin x \\ &= e^{\sin x}(\cos^2 x - \sin x). \end{aligned}$$

Differentiating again,

$$\begin{aligned} f'''(x) &= e^{\sin x}\cos x(\cos^2 x - \sin x) \\ &\quad + e^{\sin x}(-2\sin x\cos x - \cos x) \\ &= e^{\sin x}(\cos^3 x - 3\sin x\cos x - \cos x). \end{aligned}$$

Finally,

$$\begin{aligned} f^{iv}(x) &= e^{\sin x}\cos x(\cos^3 x - 3\sin x\cos x - \cos x) \\ &\quad + e^{\sin x}(-3\sin x\cos^2 x - 3\cos^2 x + 3\sin^2 x + \sin x) \\ &= e^{\sin x}\left(\cos^4 x - 6\sin x\cos^2 x - 4\cos^2 x + 3\sin^2 x + \sin x\right). \end{aligned}$$

At $x = 0$, we obtain

Function	Value at $x = 0$
f	$e^0 = 1$
f'	$e^0(1) = 1$
f''	$e^0(1-0) = 1$
f'''	$e^0(1-0-1) = 0$
f^{iv}	$e^0(1-0-4+0+0) = -3$

Substitute in Maclaurin's expansion:

$$\begin{aligned} e^{\sin x} &= 1 + x(1) + \frac{x^2}{2!}(1) + \frac{x^3}{3!}(0) + \frac{x^4}{4!}(-3) + \cdots \\ &= 1 + x + \frac{x^2}{2} - \frac{x^4}{8} + \cdots. \end{aligned}$$

Therefore,

$$\boxed{e^{\sin x} = 1 + x + \frac{x^2}{2} - \frac{x^4}{8} + \cdots.}$$